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SCHOOL OF COMMUNICATION, MULTIMEDIA & COMPUTER STUDIES

KLIR: An IoT-Based Toilet Sanitation, Analytics, and Predictive Maintenance Management System

In partial fulfillment

For the course requirements e course requirement in Bachelor of Science in Information Technology

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CHAPTER 1

INTRODUCTION

Background of the Study

Restroom sanitation is an important part of facility management because restrooms are shared spaces that are used repeatedly throughout the day. In educational institutions, the condition of these facilities affects hygiene, user comfort, public health, and the overall perception of campus services. Traditional restroom management usually depends on manual flushing, fixed cleaning schedules, and visual inspection. These methods can work for basic operations, but they do not provide a clear picture of actual usage, water consumption, device condition, or the exact time when cleaning attention is needed.

The challenge becomes more visible in high-traffic school environments. During class breaks and other peak periods, a restroom may experience a sudden increase in usage, while another restroom on the same floor may remain lightly used. If both areas follow the same fixed cleaning schedule, maintenance resources may not be assigned where they are needed most. In addition, when facility staff depend mainly on physical inspection, problems such as unusual water flow, device failure, or repeated sanitation requests may only be noticed after users have already experienced inconvenience.

The Internet of Things (IoT) provides a practical way to connect physical restroom devices with digital monitoring and facility workflows. An IoT-based sanitation system can use sensors and actuators to detect user presence, perform contactless operations, record system events, and transmit these records to a cloud platform. Koushik et al. (2017) showed that sensor-based fixtures can support more efficient water use in academic facilities. Reddy et al. (2023) also demonstrated that sanitation devices can react to sensed conditions instead of depending only on fixed schedules.

These studies support the use of sensor-driven automation as part of a more responsive restroom management approach.

Previous smart toilet studies also show the value of integrating several sanitation functions. Aher et al. (2020) proposed an IoT-based smart toilet with automated functions, while Gupta et al. (2022) emphasized contactless operation and disinfection during the COVID-19 period. Konnova et al. (2020) further discussed the importance of lid control in reducing aerosol dispersion during flushing. However, many solutions still focus on individual functions. A flushing device may not include analytics, an alerting system may not support work-order completion, and a disinfection mechanism may not be connected to occupancy-based safety checks or administrative review.

The current Klir system was developed to address this integration gap by combining restroom hardware, cloud services, a web dashboard, and a mobile custodial application into one workflow. The hardware layer is controlled by an ESP32 microcontroller connected to an ultrasonic sensor, water flow sensor, motorized lid actuator, flush valve control, and UV-C relay. The firmware follows a seven-state sequence that detects a user, opens the lid, waits for departure, closes the lid, performs a flush cycle, activates the UV-C cycle only after occupancy has cleared, and then returns to standby. This sequence reduces unnecessary physical contact and introduces a defined safety interlock between occupancy detection and disinfection control.

The system also extends beyond the toilet hardware itself. Sensor readings and device events are sent through HiveMQ Cloud using MQTT over TLS. A long-running MQTT listener processes these messages and writes structured records to Firebase Firestore. The web application presents real-time telemetry, task information, alerts, analytics, automation rules, reports, and remote controls. The mobile application connects supervisors and maintenance personnel to the same task records so that work orders can move from dispatch to acknowledgment, cleaning

execution, photo proof, quality review, and re-cleaning when necessary.

For the SDCA Annex implementation, the system maintains a directory of 22 restroom units across four floors. A separate laboratory test unit is retained for hardware diagnostics. The facility directory allows the platform to associate sensor data, cleaning tasks, assigned technicians, and reports with specific locations instead of treating the system as a single-toilet prototype. This is important because facility management decisions become more useful when administrators can compare usage and maintenance activity across multiple restroom locations.

A major improvement in the completed system is the addition of a closed-loop custodial quality assurance process. Maintenance personnel can receive tasks through the mobile application, complete a structured cleaning checklist, and submit photo evidence. Supervisors can then approve the work or flag it for recheck. The system can also support task reassignment, multi-assignee submissions, alerts for hardware problems, automated usage thresholds, continuous-flow leak detection, and hardware wear counters. These functions shift the project from simple device automation toward a more complete sanitation and maintenance management platform.

The project also recognizes that technical automation does not replace the need for responsible safety and validation. The integrated disinfection stage uses real UV-C light hardware rather than a visible-light simulation. Two supplied HP350 Spectrum Irradiance Meter reports for RZXLED UV-C modules documented peak emissions of 258.3 nm for model 05W-255NM-2b2c-N120 and 276.4 nm for model 05W-275NM-100ma-N120 at a test distance of 30 mm. The reports recorded UV-C irradiance values of 736.88 μW/cmZ and 174.77 μW/cmZ, respectively. These measurements support that the light sources produce UV-C output under the stated report conditions. However, the study does not claim measured microbial reduction in the KLIR restroom installation because no microbiological test or in-situ surface-dose validation was conducted. The

field effectiveness of the UV-C stage still depends on exposure distance, orientation, duration, enclosure design, shadowing, and surface coverage.

Through this integrated approach, Klir demonstrates how a graduating Information Technology project can combine embedded systems, networking, cloud computing, mobile development, web development, data management, cybersecurity, and software testing in one real-world application. The main contribution of the study is not a single sensor or actuator. It is the integration of physical restroom events with digital monitoring, automated rules, maintenance dispatch, supervisor review, and auditable records that can support more responsive facility management.

Statement of the Problem

Restroom sanitation in institutional environments is often managed using fixed schedules and manual observation. This creates a gap between actual restroom demand and the timing of maintenance activities. A heavily used restroom may require cleaning before the next scheduled round, while a low-traffic restroom may receive the same level of attention even when it is not necessary. Without reliable usage records, administrators have limited evidence for adjusting cleaning schedules, allocating personnel, or identifying recurring facility problems.

Another concern is the lack of an integrated connection between restroom hardware and the people responsible for maintenance. A sensor may detect usage or abnormal flow, but the information becomes less useful if it does not lead to an actionable alert, work order, or follow-up process. In the same way, a cleaning task may be completed, but without proof, timestamps, and supervisor review, it is difficult to maintain a consistent quality assurance process across multiple restroom locations.

The system therefore addresses the following practical problems: the need for touchless

occupancy-based operation; the need to monitor water flow and device events; the need for safe sequencing of lid, flush, and UV-C actions; the need for automated task generation and alerts; the need for role-based access to administrative and custodial functions; and the need for a mobile workflow that records task acknowledgment, cleaning checklist completion, photo evidence, and supervisor inspection results.

The study also addresses the limitation of treating a smart toilet as a single isolated device. A usable institutional system must support several restroom locations, maintain a clear device directory, store consistent records, and allow the same platform to be used by administrators, supervisors, technicians, and read-only viewers according to their responsibilities. The absence of this centralized workflow can result in fragmented records, slower response, duplicated tasks, and limited accountability.

Objectives of the Study

The general objective of this study is to design, develop, and validate an IoT-based toilet sanitation and maintenance management system that integrates automated restroom hardware, real-time monitoring, cloud-based task management, and mobile custodial workflows.

1. Detect toilet occupancy using an ultrasonic sensor and use the detected state as the basis for safe automated operation.
2. Implement an automated lid and flushing sequence that minimizes direct user contact with shared restroom controls.
3. Monitor water flow and restroom usage events and transmit the collected telemetry to a cloud platform for logging and analysis.
4. Implement a controlled UV-C disinfection cycle using the integrated UV-C light hardware, with activation permitted only when the system confirms that the restroom is unoccupied.

5. Develop a web-based dashboard that provides real-time telemetry, device status, alerts, analytics, maintenance records, reporting, and authorized remote controls.
6. Develop automated rules for usage-based cleaning dispatch, continuous-flow leak detection, and hardware wear monitoring. Develop a role-based maintenance workflow where administrators and supervisors can dispatch, reassign, audit, approve, or flag work orders while maintenance personnel can acknowledge and complete assigned tasks.
7. Develop a React Native mobile application that supports supervisor and maintenance roles, push notifications, task execution, checklist completion, photo proof, work history, and secure authentication.
8. Implement quality assurance features that allow supervisors to review completed tasks and require re-cleaning when submitted work does not meet the expected condition.
9. Validate the software through automated tests and functional integration checks covering the web platform, mobile application, cloud services, MQTT communication, and hardware command flow.

Scope and Limitations of the Study

1. **Occupancy Detection and Automated Sequence Module**

Scope:

- Detects user presence using an ultrasonic sensor.
- Uses a 30 cm detection threshold as the current firmware trigger.
- Runs a seven-state sequence for detection, lid movement, departure confirmation, flushing, UV-C activation, and return to standby.
- Applies timing delays to reduce false transitions and accidental activation.

Limitations:

- Ultrasonic accuracy may be affected by sensor placement, reflective surfaces, physical obstruction, and environmental conditions.
- The state timings used in the current firmware are configuration values and may require recalibration for a different restroom installation.

2. Water Flow, Usage Analytics, and Alert Module

Scope:

- Measures water-flow events and stores volume-related telemetry.
- Displays usage information and device health on the web dashboard.
- Supports automated alert rules, including continuous flow above 0.5 L/min for more than 45 seconds.
- Tracks usage counts that can be used to create custodial work orders.

Limitations:

- Analytics are limited to the measurements generated by the installed sensors and recorded system events.
- The system can provide trend-based maintenance indicators, but it does not claim advanced machine-learning prediction in the present implementation.

3. Actuation and UV-C Safety Module

Scope:

- Controls the motorized lid, flush valve, and real UV-C light module through the ESP32 relay output. Allows authorized remote commands for lid, flush, UV-C, reset, and configuration actions.
- Prevents manual UV-C activation while occupancy is detected.

- Uses confirmation controls in the web interface for safety-sensitive actions.
- Uses UV-C light modules supported by supplied HP350 spectral test reports showing peak wavelengths of 258.3 nm and 276.4 nm at a 30 mm test distance.

Limitations:

- The supplied HP350 spectral reports verify wavelength and irradiance under their stated 30 mm test setup, but the study did not perform microbiological testing of the installed KLIR system.
- Actual germicidal performance depends on wavelength, irradiance at the target surface, accumulated dose, exposure time, distance, surface coverage, shadowing, enclosure design, and operating safety.
- The current five-second UV-C cycle is a firmware timing parameter and is not presented as a validated germicidal exposure time; actual surface dose must be verified in the final installation.
- Site-specific plumbing integration may require different valves, pressure control, and installation work beyond the capstone hardware bench.

4. Web Monitoring, Dispatch, and Reporting Module

Scope:

- Provides a Next.js web dashboard for telemetry, tasks, alerts, automation rules, reports, device selection, quality review, and remote controls.
- Supports 22 restroom units across the SDCA Annex four-floor directory and one separate test unit for diagnostics.
- Uses role-based access for admin, supervisor, maintenance, and viewer accounts.
- Exports selected reports in PDF, CSV, and JSON formats.

Limitations:

- Some administrative functions require stable internet access to cloud services.
- The current implementation is designed for the SDCA Annex directory and would require configuration changes when deployed to another institution.

5. Mobile Custodial and Supervisor Module

Scope:

- Supports supervisor and maintenance roles.
- Delivers FCM task notifications and real-time Firestore synchronization.
- Provides a three-step technician task flow with arrival proof, a 10-item checklist, and completion proof.
- Supports additional area photos, timestamp/location overlays, task reassignment, multi-assignee submissions, work history, and supervisor review.
- Uses biometric authentication where available, with a password fallback for gated supervisor review. Queues select task actions when connectivity is interrupted and synchronizes them when the connection returns.

Limitations:

- Native push notifications should be validated on a physical Android device with Google Play Services.
- Biometric functions depend on compatible hardware and enrollment in the device operating system.
- The mobile build is intended for controlled institutional deployment and testing; public app-store distribution is outside the present capstone scope.

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6. Quality Assurance and Security Module

Scope:

- Requires photo evidence and checklist completion for custodial submissions.
- Allows supervisors to approve completed work or flag it for recheck.
- Uses Firebase Authentication, server-side token verification, and role-based access control.
- Protects critical configuration changes with re-authentication and limits actuator controls to authorized roles.

Limitations:

- The system records evidence of cleaning work but does not independently measure actual surface cleanliness or microbial reduction.
- Security depends on proper protection of credentials, environment variables, Firebase rules, and authorized user accounts.

Conceptual Framework

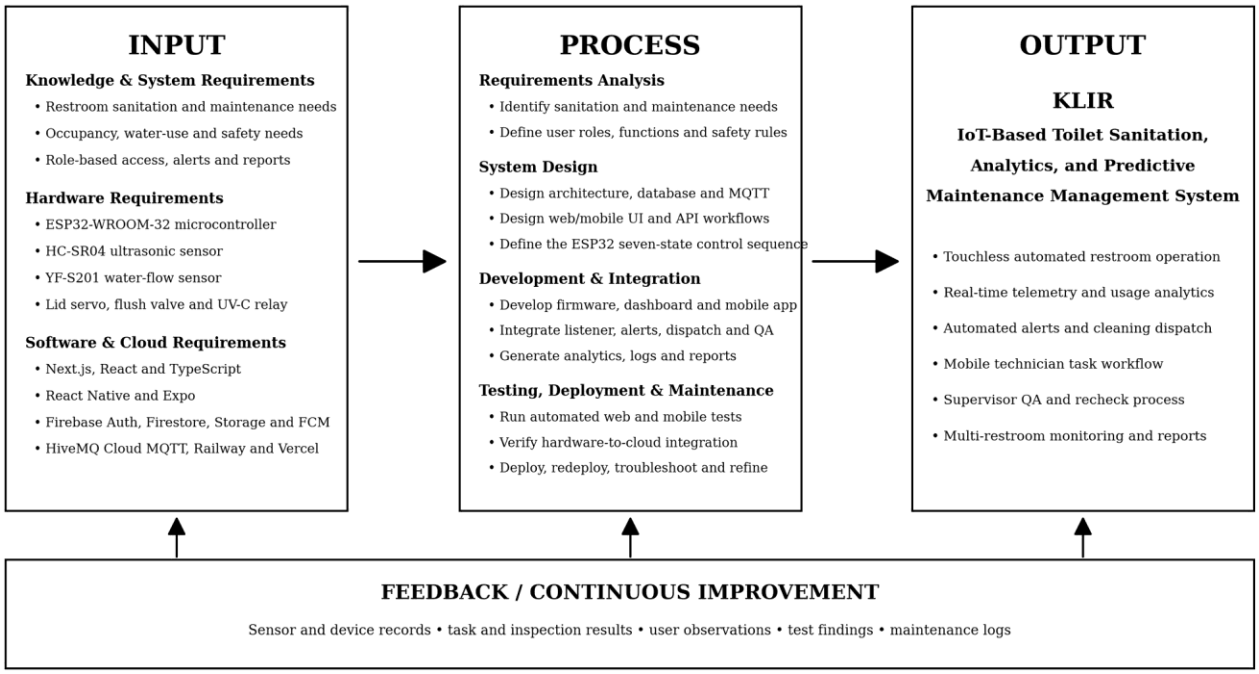


Figure 2. Input-Process-Output (IPO) conceptual framework of the KLIR system with a feedback loop for continuous system improvement.

The conceptual framework of the study follows the Input-Process-Output (IPO) model with a feedback loop. The input includes the knowledge, system requirements, hardware components, software tools, and cloud services needed to develop KLIR. These inputs are used during the process stage, which covers requirements analysis, system design, development and integration, testing, deployment, and maintenance. The process produces the KLIR IoT-Based Toilet Sanitation, Analytics, and Predictive Maintenance Management System, which provides touchless restroom automation, real-time monitoring, maintenance dispatch, mobile task execution, supervisor quality assurance, and multi-restroom reporting. The feedback loop represents the continuous use of sensor records, task and inspection results, user observations, testing findings,

and maintenance logs to identify issues and improve the system as development and operational testing continue.

Significance of the Study

- **Facility Managers and Administrators.** The system provides a centralized view of restroom usage, alerts, device status, work orders, and reports that can support more evidence-based facility decisions.
- **Supervisors.** The supervisor workflow provides live personnel status, task dispatch and reassignment, completed-work review, approval, and recheck functions.
- **Maintenance Personnel.** The mobile application gives technicians a structured way to receive tasks, acknowledge work, follow a checklist, submit photo proof, and maintain a work history.
- **Restroom Users.** Touchless operation and faster maintenance response can reduce direct contact with shared controls and improve the consistency of facility servicing.
- **St. Dominic College of Asia.** The project demonstrates how IoT and software systems can be applied to campus operations using a multi-floor restroom directory and centralized cloud records.
- **Future Researchers and Developers.** The project can serve as a reference for studies involving smart sanitation, connected facilities, custodial dispatch, IoT safety interlocks, and mobile quality assurance.
- **Information Technology Students and Professionals.** The project demonstrates the integration of software engineering, mobile and web development, cloud computing, networking, database management, cybersecurity, automated testing, and embedded systems.

Alignment of the Study to the Information Technology (IT) Major

The project aligns with the Bachelor of Science in Information Technology program because the main work of the study is the design and integration of an information system that connects hardware, software, cloud services, users, and operational processes. The ESP32 and sensors serve as the physical data source, while the larger project involves application development, secure communication, database design, real-time synchronization, testing, deployment, and system maintenance.

- **Systems Development.** The project follows an iterative Agile development process and integrates requirements, design, implementation, testing, deployment, and maintenance.
- **Software Engineering.** The system uses typed application code, REST API routes, reusable services, validation, automated tests, and separated web/mobile codebases.
- **Networking and IoT.** MQTT over TLS connects the ESP32 devices with HiveMQ Cloud and the backend listener.
- **Database Management.** Firestore stores devices, sensor readings, tasks, alerts, automation rules, maintenance notes, and user-related records.
- **Cybersecurity.** Authentication, role-based access control, protected server routes, re-authentication, and controlled actuator permissions are part of the system architecture.
- **Cloud Computing.** The application uses Firebase, Vercel, Railway, HiveMQ Cloud, and FCM as coordinated cloud services.
- **Data Analytics.** Water use, flush activity, alerts, device health, and task records are summarized for operational decisions.
- **Human-Computer Interaction.** The web interface follows responsive and accessibility-oriented design, while the mobile application provides role-specific workflows for

supervisors and technicians.

Definition of Terms

Technical Terms

- **Automated Rules Engine -** A set of configurable conditions that generates alerts or work orders based on usage counts, continuous flow, or other device events.
- **Cloud Firestore -** The Firebase NoSQL database used by the system for real-time device, task, alert, configuration, and maintenance records.
- **Firebase Cloud Messaging (FCM) -** The notification service used to deliver work-order and system alerts to registered mobile devices.
- **Hardware Interlock -** A safety rule that prevents an actuator from running when a required condition is not satisfied, such as blocking UV-C activation while occupancy is detected.
- **HiveMQ Cloud -** The cloud MQTT broker used to route telemetry and command messages between the ESP32 and backend services over TLS.
- **MQTT -** A lightweight publish-subscribe messaging protocol used for IoT telemetry and actuator commands.
- **MQTT Listener -** The long-running backend service that subscribes to device topics, validates events, writes Firestore records, evaluates alerts, and sends notifications.
- **Role-Based Access Control (RBAC) -** A security method that grants functions according to the authenticated user role, including admin, supervisor, maintenance, and viewer.
- **Supervisor QA Audit -** The review process where a supervisor approves a completed cleaning task or flags it for recheck.

- **SWR Cache -** A client-side data strategy used by the mobile application to show cached task information immediately while refreshing data in the background.
- Telemetry - Sensor and device information transmitted by the IoT hardware, including distance, flow, flush, lid, UV-C, and connectivity events.
- **ToiletUnitSelect -** The searchable web control used to locate restroom units by floor, gender, or room code.
- **UV-C -** Ultraviolet-C radiation used by the system as a controlled disinfection actuator. KLIR uses real UV-C light hardware, with supplied HP350 spectrum irradiance reports showing spectral peaks at 258.3 nm and 276.4 nm. The reports verify UV-C emission under the stated test conditions but do not independently establish microbial reduction in the installed restroom system.
- **WCAG 2.2 AA -** An accessibility target applied to the web interface for contrast, keyboard interaction, responsive reflow, and accessible naming of interactive elements.
- **Work Order -** A cleaning or maintenance task assigned to one or more custodial personnel and tracked from dispatch to completion and quality review.

Non-Technical Terms

- **Occupancy Detection -** The process of determining whether a user is present near the toilet using a distance sensor.
- **Manual Override -** An authorized action that allows an administrator or supervisor to operate selected hardware controls from the web dashboard.
- **Photo Proof -** An image submitted by maintenance personnel to document the condition of the restroom before or after cleaning.
- **Recheck -** A supervisor decision requiring the assigned technician to repeat or correct

a cleaning task.

- **Usage Analytics -** Summaries and trends based on recorded restroom use, water flow, device events, alerts, and task activity.

CHAPTER 2

Review of Related Literature and Studies

2.1. Ultraviolet Germicidal Irradiation (UVGI) for Surface Disinfection

2.1.1. Systematic Review on UVGI Effectiveness

Oringanje et al. (2025) conducted a systematic review showing that ultraviolet germicidal irradiation, when used together with standard cleaning methods, can significantly reduce viral contamination on surfaces. The review reported effective short exposures at 254 nm and 222 nm, but also noted that the number of controlled trials was limited and that patient outcome evidence was not yet clear. The finding supports UV-C as a possible additional sanitation method but also shows the importance of controlled implementation and validation.

2.1.2. Experimental Study on Viral Reduction

Rudhart et al. (2022) evaluated UV-C surface disinfection using a bacteriophage model and reported up to a 5.9 log10 reduction after 75 seconds of exposure under the tested conditions. This indicates that UV-C can be highly effective when the wavelength, dose, exposure time, and surface conditions are properly controlled.

2.1.3. UV-C Inactivation of SARS-CoV-2

Biasin et al. (2021) investigated UV-C irradiation against SARS-CoV-2 under laboratory conditions. The study reported strong inactivation at relatively low doses and complete inactivation at the highest tested dose. These findings are relevant to the design of disinfection systems, but their direct application to a restroom environment still depends on factors such as distance, surface coverage, shadowing, enclosure design, and exposure safety. For this reason, the present capstone treats UV-C as a controlled hardware feature and does not claim verified microbial reduction without separate microbiological validation.

2.1.4. Spectral Test Documentation of the UV-C Modules Used in KLIR

To support the actual UV-C hardware implementation, the project reviewed two supplied HP350 Spectrum Irradiance Meter reports for RZXLED light modules. The first report, dated April 1, 2022, for model 05W-255NM-2b2c-N120 recorded a peak wavelength of 258.3 nm, a half width of 12.6 nm, total irradiance (Ee) of 745.97 µW/cmZ, and UV-C irradiance of 736.88 µW/cmZ at a distance of 30 mm. The second report, dated April 2, 2022, for model 05W-275NM-100ma-N120 recorded a peak wavelength of 276.4 nm, a half width of 11.0 nm, total irradiance of 248.97 µW/cmZ, and UV-C irradiance of 174.77 µW/cmZ at the same distance. These reports provide component-level evidence that the selected light sources emit UV-C radiation under the stated test conditions (RZXLED, 2022a; RZXLED, 2022b). They do not replace microbiological testing or in-situ irradiance measurements at the actual toilet surfaces; therefore, they are used in this study as hardware spectral verification rather than proof of sanitation effectiveness.

2.2. Hygiene Disparities Between Public and Private Restrooms

2.2.1. Microbiological Contamination in Public Restrooms

Public restrooms generally experience higher contamination risk because of repeated contact with shared surfaces and high traffic. Even when cleaning is scheduled regularly, flush controls, door hardware, fixtures, and nearby surfaces may become contaminated again shortly after cleaning. This supports the need for reduced-touch operation and maintenance processes that can respond to actual demand.

2.2.2. Hospital Toilet Sanitation Assessments

Amin et al. (2024) reported that only 33% of government hospital toilets and 56% of private hospital toilets in their Dhaka assessment met acceptable cleanliness standards. High user volume was identified as an important sanitation challenge. Although hospitals differ from educational facilities, the findings show how user density can affect the ability of fixed cleaning schedules to maintain restroom conditions.

2.2.3. Ventilation and Airborne Contamination

Lee and Tham (2021) discussed cross-infection risk in public toilets with insufficient ventilation. Their findings indicate that sanitation should not be viewed only as surface cleaning because airflow, flushing behavior, and enclosure conditions also influence hygiene risk.

2.2.4. User Perception and Public Health Impact

Restroom cleanliness affects whether users feel comfortable using a facility. A poorly maintained restroom may cause users to avoid the area or change normal hygiene behavior. This makes restroom sanitation both an operational concern and a service-quality

issue for institutions.

2.2.5. Maintenance Interventions and Sanitation Improvement

Structured maintenance, regular monitoring, and clear responsibility improve the usability of shared restrooms. For smart facilities, the value of monitoring increases when detected conditions can generate an actual maintenance response instead of only appearing as dashboard information.

2.3. Hands-Free and Smart Toilet Technologies

2.3.1. Limitations of Traditional Toilet Systems

Traditional toilet systems often require users to touch shared flush controls. Repeated contact can increase cross-contamination risk, while flushing may also produce aerosols. These limitations support the use of touchless controls and automatic sequencing in high-use restrooms.

2.3.2. Sensor-Activated Flush Systems

Sensor-activated flush systems reduce the need for direct contact and can produce consistent activation behavior. When combined with water-flow measurement and event logging, the flush event can also become a useful source of usage information for maintenance planning.

2.3.3. Smart Sanitation Features

Smart sanitation systems can combine occupancy sensing, automatic flushing, disinfection control, water monitoring, alerting, and maintenance records. The value of the technology is stronger when these features share a common data platform and can trigger a coordinated response.

2.3.4. Evidence-Based Design Considerations

A smart sanitation system should distinguish between what the hardware can perform and what has been scientifically verified. Sensor reliability, user response, water consumption, software performance, and task completion can be validated through engineering tests, while claims about microbial reduction require a separate laboratory or field microbiology method.

2.4. Mobile Notification Systems in IoT-Based Facility Management

2.4.1. Role of Push Notifications in Maintenance Response Systems

IoT-based facility management systems increasingly use mobile notifications to connect automated events with human response. See-To et al. (2023) discussed proactive maintenance management for IoT-enabled smart toilets, supporting the idea that connected maintenance tools can improve response efficiency. In the present system, notifications are tied to work orders so that an alert can continue into acknowledgment, task execution, and quality review.

2.4.2. Firebase Cloud Messaging for IoT Alert Delivery

Firebase Cloud Messaging provides server-to-device push notification delivery for mobile applications. It is suitable for the project because it integrates with Firebase Authentication and Firestore, allowing the same user and task records to support both real-time application data and mobile alerts. Mokar et al. (2019) also demonstrated the use of FCM for controlling and communicating with mobile applications.

2.4.3. React Native for Cross-Platform IoT Companion Applications

React Native allows a mobile application to be developed from one JavaScript or TypeScript codebase while still using native device functions. In this project, it supports

the supervisor and technician interfaces, biometric authentication, image processing, camera-based evidence, local storage, and integration with Firebase services. This reduces the need to maintain separate Android and iOS application logic during capstone development.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter presents the project design, system framework, feasibility assessment, development process, testing approach, deployment architecture, and maintenance plan used for the current Klir implementation. The methodology reflects the completed web, mobile, cloud, and hardware functions rather than an earlier single-device prototype.

A. PROJECT DESIGN

a.1. System Objective

The primary system objective was to develop an IoT-based sanitation and custodial management platform that can automate selected toilet functions, record operational data, and connect those events to facility personnel. The project was designed so that the toilet hardware could continue to perform its local sequence while the cloud platform provides monitoring, reporting, alerting, task dispatch, and remote administrative functions.

At the hardware level, the system uses an ESP32-WROOM-32, an ultrasonic distance sensor, a YF-S201 water flow sensor, a servo for the lid, a controlled flush output, and a relay-controlled UV-C light module. The UV-C source is supported by supplied HP350 spectrum irradiance test reports that document UV-C spectral output for the selected RZXLED modules. The firmware follows a non-blocking seven-state finite state machine to reduce timing conflicts and make the system behavior easier to test. The local

logic confirms departure before lid closing, flushing, and UV-C activation. This allows the most safety-sensitive sequence to remain at the edge device instead of depending completely on a cloud command.

At the application level, the web system provides a central operations view for device telemetry, restroom selection, task dispatch, quality assurance, automated rules, alerts, reports, and hardware controls. The mobile application extends the workflow to supervisors and custodial personnel. Supervisors can monitor staff status, dispatch or reassign tasks, and review completed work, while technicians can receive tasks, complete a structured checklist, attach photo evidence, and submit their work for approval.

The final system objective therefore includes both automation and accountability. A cleaning requirement is not considered complete simply because an alert was generated. The task is tracked through assignment, acknowledgment, execution, submission, and supervisor review so that the system produces a usable maintenance record.

a.2. System Framework

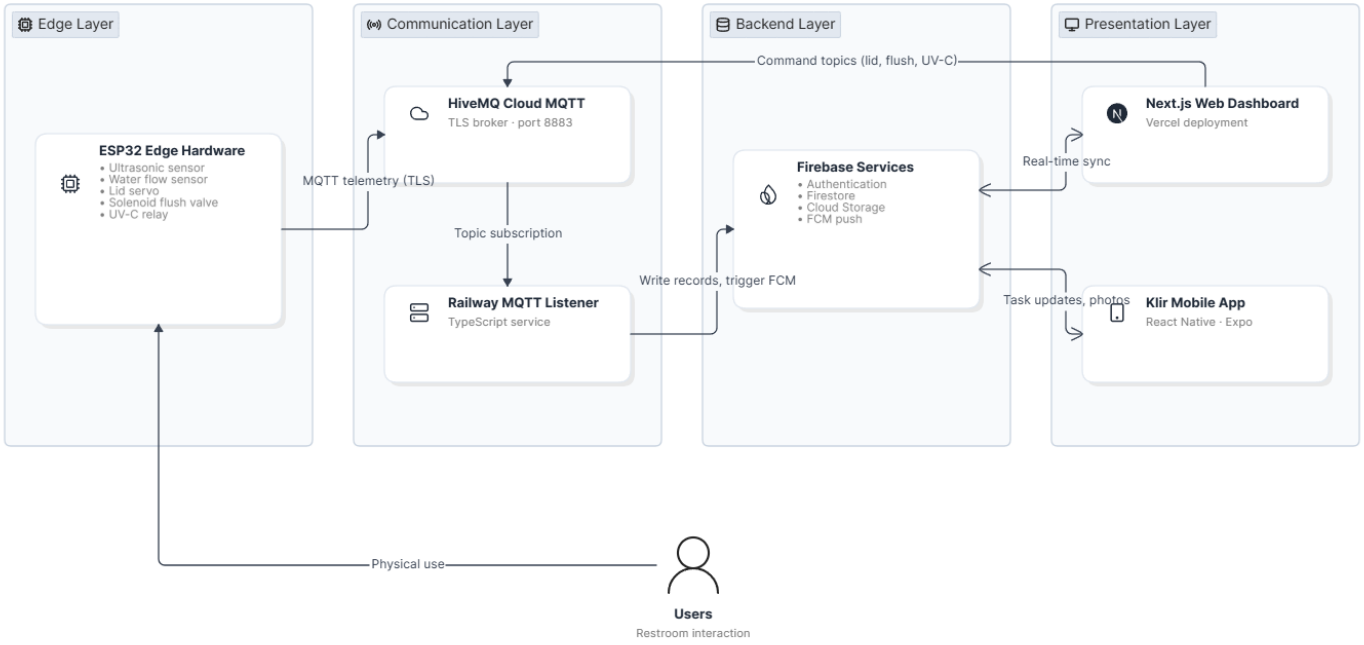


Figure 1. End-to-end system architecture of Klir.

The architecture is divided into an edge layer, MQTT communication layer, ingestion and rules layer, Firebase cloud layer, web presentation layer, and mobile application layer. This separation was used because the ESP32 must perform real-time local control, while the web and mobile applications require cloud-hosted data that can be accessed by several users.

The edge layer publishes sensor and event messages to HiveMQ Cloud using MQTT with TLS. The main telemetry topics include ultrasonic distance, water-flow events, lid events, flush events, and UV-C events. Command topics allow authorized web functions to request flush, UV-C, lid open or close, configuration updates, and device reset. QoS 1 is used for the documented telemetry topics so that the broker provides at-least-once delivery behavior.

A standalone TypeScript MQTT listener runs as a long-lived service. It subscribes to sensor

and event topics, validates incoming payloads, assigns timestamps, writes structured data to Firestore, evaluates automation rules, tracks hardware counters, and triggers FCM notifications when needed. This service is hosted separately from the serverless web application because a persistent MQTT connection is not suitable for short-lived serverless execution.

Firebase provides authentication, Firestore persistence, storage for mobile proof images, and cloud messaging. The web application uses server-side token verification and role checking for protected API routes. The mobile application uses the same authenticated user records and task collections so that task state remains synchronized between the supervisor view and technician workspace. The current facility directory represents 22 restroom units and 96 individual fixtures across the first to fourth floors of the SDCA Annex Building. The first floor contains four registered restrooms, while the second, third, and fourth floors each contain six, including a dedicated PWD (Left Wing) and PWD (Right Wing) restroom per floor.

Hardware and Firmware Design

GPIO	Function	Current Use
12	Ultrasonic TRIG	Distance measurement trigger
13	Ultrasonic ECHO	Distance measurement return with voltage-divider protection
14	Flush output	Controls the 12 V solenoid flush valve through a relay
27	UV-C relay	Controls the real UV-C light module through the relay output
25	Lid servo	PWM control for motorized lid movement
32	Water flow	Interrupt-based pulse input from YF-S201

2	Status LED	Connectivity and operating-state indication

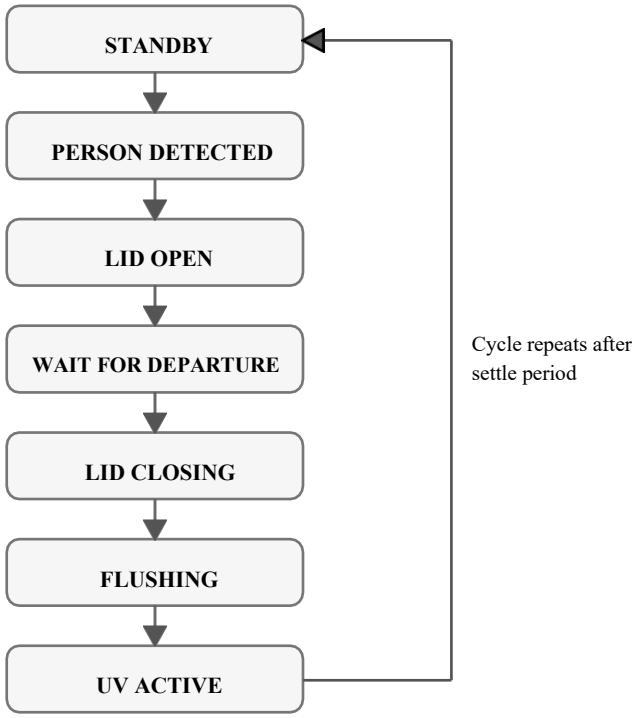


Figure 2. Seven-state firmware sequence used by the ESP32.

The default detection threshold is 30 cm. After the lid opens, a five-second grace period is used before the system begins departure confirmation. The user must remain absent for three seconds before the lid-closing state continues. The current flush duration is three seconds, followed by a five-second UV-C control cycle, then a two-second standby settle period. These values are implementation parameters and can be recalibrated through configuration when the installation

changes. The five-second UV-C timing demonstrates the automated control sequence and occupancy interlock; it is not treated as a validated germicidal dose because irradiance at the actual toilet surfaces was not measured as part of this study.

The hardware also follows a power-isolation rule. High-current devices such as the servo and valve are not powered directly from the ESP32 logic rail. The servo uses an external regulated 5 V supply, with a common ground and a large decoupling capacitor near the load. The ultrasonic echo line is reduced to the ESP32-safe logic level using a resistor divider. These measures reduce resets, unstable readings, and GPIO overvoltage risk.

UV-C Light Hardware and Spectral Verification

The UV-C stage was implemented using real UV-C light hardware controlled by the ESP32 through the UV-C relay. To document the optical characteristics of the selected light sources, the project reviewed two supplied HP350 Spectrum Irradiance Meter test reports from RZXLED. Both reports were taken at a distance of 30 mm and at 25°C and 65% relative humidity. Table 1 summarizes the reported measurements. The values confirm UV-C spectral output from the modules under the stated test setup, but they are not used as a direct measurement of the irradiance received by toilet surfaces in the KLIR installation.

Parameter	Report 1	Report 2
RZXLED model	05W-255NM-2b2c-N120	05W-275NM-100ma-N120
Test date	April 1, 2022	April 2, 2022
Test distance	30 mm	30 mm
Peak wavelength	258.3 nm	276.4 nm

Half width	12.6 nm	11.0 nm
Total irradiance (Ee)	745.97 µW/cmZ	248.97 µW/cmZ
UVA	2.63 µW/cmZ	7.23 µW/cmZ
UVB	6.47 µW/cmZ	66.96 µW/cmZ
UV-C	736.88 µW/cmZ	174.77 µW/cmZ

Table 1. Summary of the supplied HP350 spectral irradiance test reports for the UV-C light modules used in KLIR.

The first module produced a reported spectral peak at 258.3 nm with 736.88 µW/cmZ attributed to UV-C, while the second module peaked at 276.4 nm with 174.77 µW/cmZ attributed to UV-C. These results support the replacement of the earlier simulated light concept with actual UV-C hardware. However, because the reports describe a 30 mm test arrangement rather than the installed toilet geometry, the project does not convert these values into a claimed sanitation dose or microbial-reduction percentage. In-situ irradiance mapping and microbiological validation are recommended before defining a final disinfection exposure time for operational deployment.

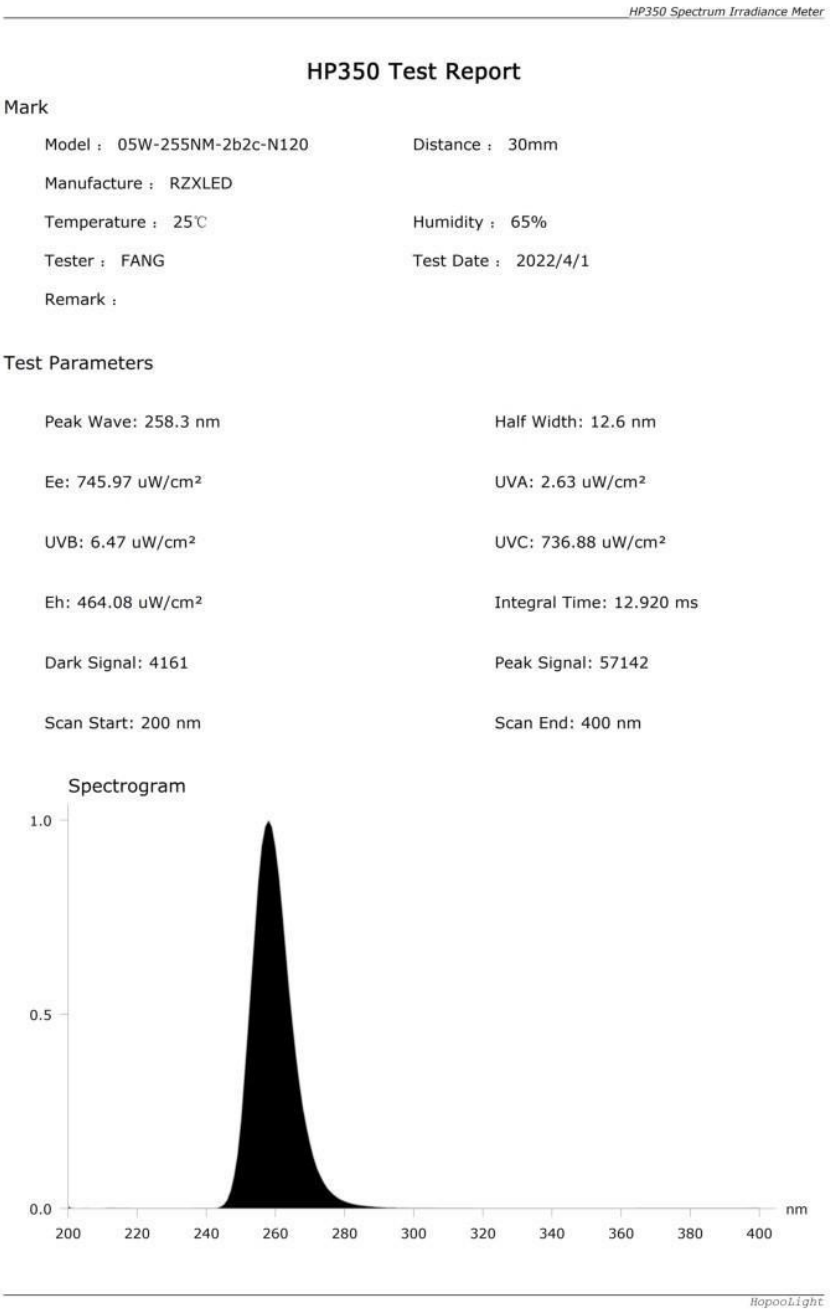
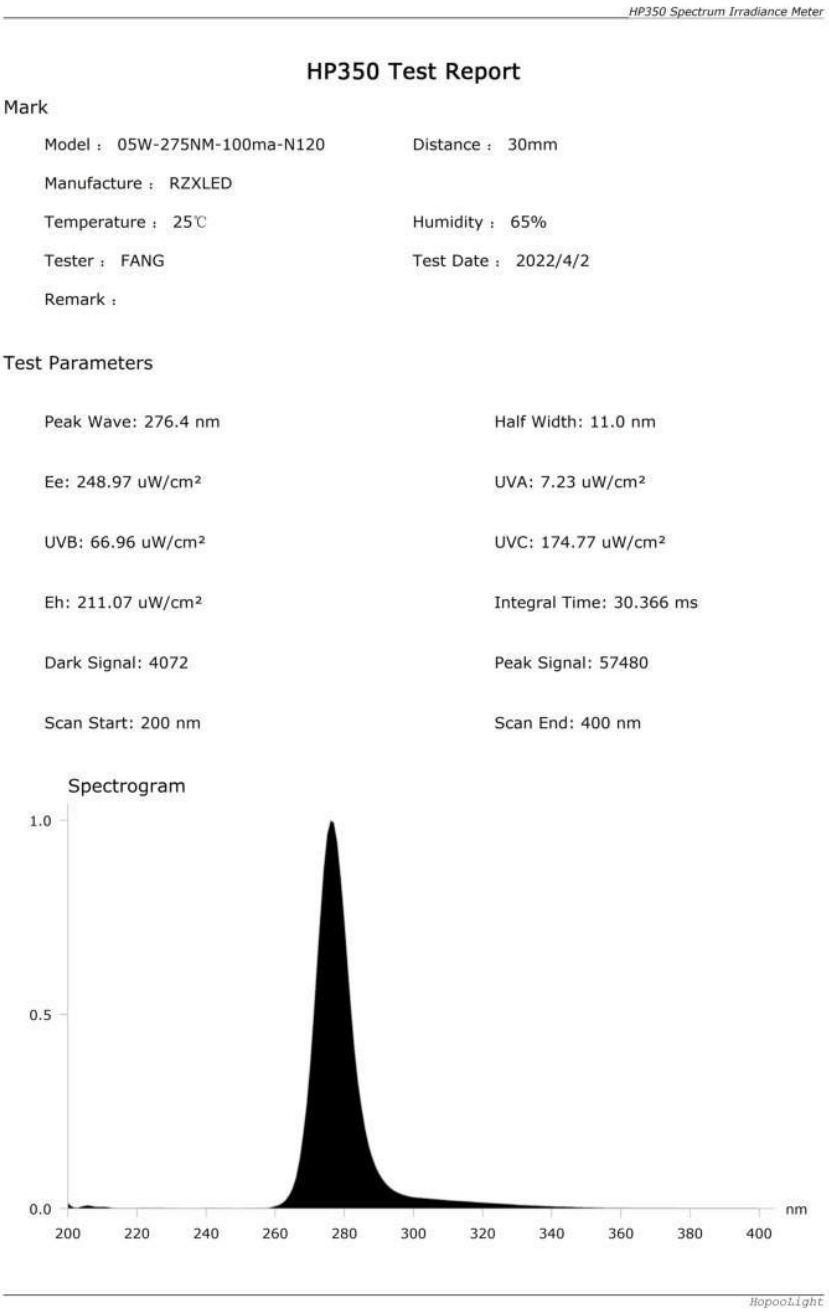


Figure 3. HP350 spectrum irradiance test report for RZXLED model 05W-255NM-2b2c-N120, showing a reported peak wavelength of 258.3 nm. Source: RZXLED (2022a).



HP350 Spectrum Irradiance Meter

HP350 Test Report

Mark

Model： 05W-275NM-100ma-N120 Distance： 30mm

Manufacture： RZXLED

Temperature： 25℃ Humidity： 65%

Tester： FANG Test Date： 2022/4/2

Remark：

Test Parameters

Peak Wave: 276.4 nm Half Width: 11.0 nm

Ee: 248.97 uW/cm² UVA: 7.23 uW/cm²

UVB: 66.96 uW/cm² UVC: 174.77 uW/cm²

Eh: 211.07 uW/cm² Integral Time: 30.366 ms

Dark Signal: 4072 Peak Signal: 57480

Scan Start: 200 nm Scan End: 400 nm

Spectrogram

HopooLight

Figure 4. HP350 spectrum irradiance test report for RZXLED model 05W-275NM-100ma-N120, showing a reported peak wavelength of 276.4 nm. Source: RZXLED (2022b).

MQTT Event and Command Structure

Category	Topic / Example	Purpose
Sensor	toilet/sensors/ultrasonic	Distance telemetry
Sensor	toilet/sensors/waterflow	Water volume and duration
Event	toilet/events/lid	Lid open/close event
Event	toilet/events/pump	Flush state event
Event	toilet/events/uv	UV-C cycle event
Command	toilet/commands/pump	Manual flush on/off
Command	toilet/commands/uv	Manual UV-C on/off
Command	toilet/commands/lid	Open/close lid
Command	toilet/commands/config	Update supported timing or thresholds
Command	toilet/commands/reset	Device software reset

a.3. Feasibility Study

a.3.1. Technical Feasibility

The system is technically feasible because the major hardware and software components use established interfaces and widely available development tools. The ESP32 provides integrated Wi-Fi, digital input/output pins, PWM capability, and enough processing power for the local state machine. The sensor and actuator modules can be tested independently before they are connected to the full sanitation sequence.

The web platform uses Next.js 16.1, React 18.3, TypeScript, Tailwind CSS, DaisyUI, Firebase Admin, and an MQTT listener service. The mobile platform uses

React Native 0.81.5 with Expo SDK 54 and TypeScript 5.9. The mobile application also uses libraries for local authentication, image manipulation, local cache storage, and image watermark capture. These tools are well documented and can be maintained by an IT development team with experience in JavaScript/TypeScript and Firebase.

The cloud architecture is also feasible because each service has a clear responsibility. Vercel hosts the web dashboard and API routes, Railway hosts the long-running MQTT listener, HiveMQ Cloud handles publish-subscribe messaging, and Firebase provides shared application data and identity services. The separation reduces the risk of trying to keep a permanent MQTT connection inside a serverless runtime.

Technical risks remain in the areas of network interruption, moisture exposure, actuator power stability, cloud service quotas, and hardware calibration. The design mitigates these risks through local edge control, separate power rails, reconnect logic, device heartbeat monitoring, role-limited remote commands, and a mobile offline queue for selected task actions.

The real UV-C hardware is also technically supported by the supplied HP350 spectral reports. The reports document UV-C output for the selected RZXLED modules at a 30 mm measurement distance. This improves confidence that the system is controlling an actual UV-C source rather than a visual indicator. Nevertheless, deployment feasibility still requires enclosure safety, protection against human exposure, and site-specific irradiance or dose verification before the UV-C cycle can be treated as a validated disinfection process.

a.3.2. Operational Feasibility

Operationally, the system is designed around the responsibilities already present in facility management. Administrators and supervisors use the web dashboard to monitor devices, configure rules, create tasks, review alerts, and approve completed work. Maintenance personnel use a mobile workspace focused on task execution. This role separation reduces the amount of training required because each user is presented mainly with the functions related to their job.

The mobile workflow is designed for field use. A technician receives a task, provides arrival evidence, completes the required checklist, and submits completion evidence. The application can compress proof images before upload and can store selected pending operations locally if connectivity is interrupted. Supervisors can see whether personnel are available, on task, or offline, making reassignment more practical during actual custodial operations.

The quality assurance process also improves operational feasibility because completed work does not immediately disappear into a history list. A supervisor can approve the submission or flag it for recheck, and the task record preserves the review result. This creates a clearer feedback loop between the personnel who perform the cleaning and the staff who are responsible for verifying facility condition.

a.3.3. Economic Feasibility

The original prototype cost analysis estimated approximately PHP 3,500 for one development unit using an ESP32, sensors, servo, pump/flush component, UV lighting component, power supply, wiring, and miscellaneous materials. This

estimate remains useful as a prototype baseline, but it should not be treated as the cost of full deployment across all 22 restroom locations. The current system includes additional cloud, mobile, safety, storage, and multi-location functions that can increase installation and operating cost depending on hardware quality and institutional requirements.

Component	**Prototype Estimate (PHP)**
ESP32 microcontroller	350
Ultrasonic sensor	200
Water-flow sensor	250
Servo motor	300
Flush / water actuation component	500
UV-C light module	700
Power supply and wiring	400
Miscellaneous materials	800
Total prototype baseline	3,500

Most development software can be used without license fees for a student project. Firebase, HiveMQ Cloud, Vercel, Railway, and related services may provide free or low-cost tiers during development, but a larger institutional deployment may require paid plans as message volume, storage, bandwidth, or uptime requirements increase. Certified UV-C hardware, waterproof enclosures, electrical protection, plumbing installation, and maintenance labor are also not included in the original PHP 3,500 prototype estimate.

a.3.4. Cost Estimation for Full-Scale Production

During the CP2 progress evaluation, the panel observed that the current prototype operates as a retrofit sensor and automation module installed onto an existing toilet fixture, rather than a complete, purpose-built toilet unit manufactured from the ground up. Constructing a full production-grade unit was determined to fall outside the practical scope and resources available for this capstone. In place of a physical build, the panel recommended that the study instead produce a cost-estimation analysis projecting what a fully manufactured, market-ready version of the system would require, covering component sourcing, manufacturing processes, regulatory compliance, and estimated per-unit cost across different production volumes. This section presents that analysis as a supplementary feasibility study; it does not describe the system that was implemented and tested for this capstone.

Projected System Architecture at Production Scale

A production version of Klir would replace several of the prototype's retrofit components with industrial-grade equivalents suited to mass manufacturing and long-term reliability. Presence detection would shift from the current ultrasonic sensor to a solid-state millimeter-wave radar module, offering more consistent detection through varying materials and mounting positions. Lid and seat actuation would use a pair of high-torque DC gearmotors rather than a small hobby servo, since a full ceramic lid and seat assembly requires substantially more lifting force than the prototype's lightweight test rig. Flush actuation would use a linear solenoid valve sized for standard plumbing pressure, and the UV-C disinfection stage would use an array of dedicated deep-UV LEDs rather than the single relay-driven strip used in the prototype. All electronics would run on a medical-grade isolated power supply with ground-fault protection, reflecting the safety

requirements of a consumer bathroom fixture rather than a lab bench setup.

Manufacturing Process Overview

A production unit would be manufactured in two parallel tracks that are later joined into a single assembly. The ceramic toilet bowl would go through a standard vitreous china process: clay slurry preparation, high-pressure slip casting into a bowl mold, drying and glaze application, high-temperature kiln firing, and a final pressure and leak test. In parallel, the electronic seat and control housing would be injection-molded from antibacterial-treated plastic, with the internal circuit boards assembled, coated for moisture resistance, and sealed into the housing through ultrasonic welding. Once both halves are complete, the ceramic base and the electronic seat assembly would be joined and undergo a final full-unit safety and functionality test before packaging.

Estimated Cost by Production Phase

The cost-estimation study projects three distinct phases between the current prototype stage and a fully commercial product, summarized below:

Phase	**Estimated Unit Cost**	**One-Time Tooling / Setup Cost**	**Estimated Time to First Unit**
Proof-of-Concept (Capstone-equivalent stage)	₱10,600 – ₱15,700 (approx. $190–$280)	None (3D-printed parts, bench prototyping)	2–4 weeks
Pilot Run (100 units, soft tooling)	₱7,000 – ₱8,700 (approx. $125–$155)	₱450,000 – ₱840,000 (approx. $8,000–$15,000)	2–3 months

Mass Production (10,000+ units, hard tooling)	₱3,800 – ₱4,600 (approx. $68–$82)	₱4.8M – ₱7.8M (approx. $85,000–$140,000)	6–9 months

The unit cost decreases significantly at higher volumes primarily because tooling costs such as steel injection molds and ceramic casting molds are one-time investments that get distributed across a much larger number of units, and because bulk component pricing from suppliers drops sharply once order quantities reach the thousands.

Primary Cost Drivers

Several factors account for the difference in cost between the current prototype and a production-grade unit. The ceramic bowl itself, sourced as a standard manufactured toilet fixture, represents a fixed baseline cost that the prototype's retrofit approach avoids entirely by reusing an existing installed toilet. Industrial-grade sensors and actuators particularly the high-torque gearmotors and the radar presence sensor cost more per unit than the hobbyist-grade components used in the prototype, though their price drops considerably at bulk order volumes. Regulatory certification and safety testing, required before any such product could be legally sold, also represents a meaningful one-time cost that a campus prototype is not required to undergo.

Regulatory Standards Applicable to a Production Unit

A commercially manufactured version of the system would need to comply with several established safety standards before it could be marketed. Electrical safety would need to satisfy household appliance standards covering insulation class and ground-fault protection. Structural requirements would need to confirm that the seat and lid assembly can support standard static load

requirements without failure. Because the system includes UV-C disinfection, it would additionally need to meet photobiological safety limits governing safe UV exposure thresholds, with a hardware-level safety interlock ensuring the UV-C module cannot activate while the lid is open or a user is present.

Scope Clarification

This cost-estimation study is presented as a forward-looking feasibility exercise requested by the panel in place of a full production build, and reflects a hypothetical fully manufactured version of the system rather than the prototype actually developed, deployed, and evaluated at the SDCA Annex Building. All testing, evaluation results, and conclusions presented elsewhere in this study pertain solely to the retrofit prototype described in the preceding sections of this chapter.

a.4. System Design - System Development Life Cycle

The project followed an Agile development approach because the system contained several parts that had to be improved together. Changes to the hardware event format affected the backend listener, Firestore records, dashboard components, and sometimes the mobile application. An iterative process allowed the team to test one feature, identify integration problems, and update the related modules before moving to the next set of functions.

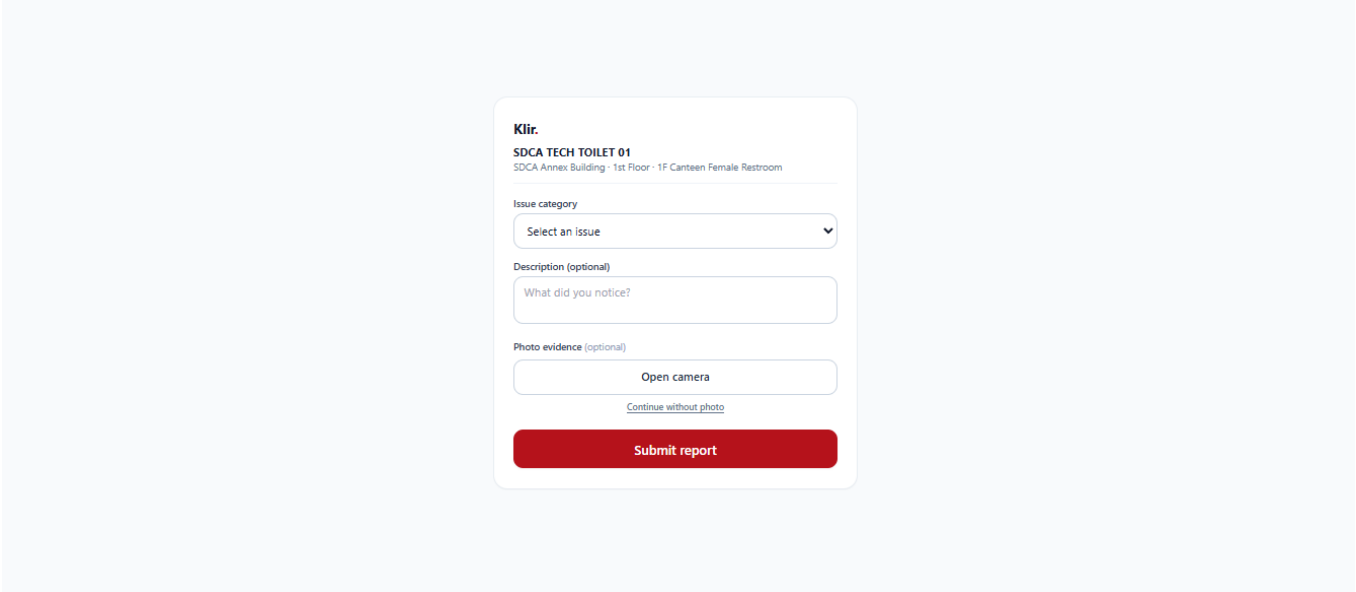
The development work was divided into parallel areas: ESP32 hardware and firmware, web dashboard and API development, cloud and MQTT integration, and mobile application development. Integration checkpoints were used to verify that data produced by one part of the system could be correctly understood by the next part. Automated tests were later added to the web and mobile codebases to reduce regression during feature expansion.

Phase	Main Activities
1. Planning and Requirements	Define target restrooms, users, sensor functions, safety rules, data records, and application roles.
2. Architecture and Data Design	Define MQTT topics, Firestore collections, API routes, task states, device directory, and role permissions.
3. Hardware and Edge Development	Implement sensor reading, actuator control, seven-state FSM, connectivity, and MQTT publishing.
4. Web Development	Implement authentication, dashboard, device selection, controls, alerts, automation rules, tasks, QA review, reports, and accessibility.
5. Mobile Development	Implement supervisor and technician role routing, task inbox, reassignment, three-step execution, photo proof, biometrics, offline cache, and history.
6. Integration and Testing	Connect MQTT, Firestore, FCM, web, mobile, and hardware; execute automated tests and functional checks.
7. Deployment and Maintenance	Deploy web and listener services, configure Firebase/HiveMQ, build mobile client, monitor logs, and maintain stable versions.

a.4.1. User Interface

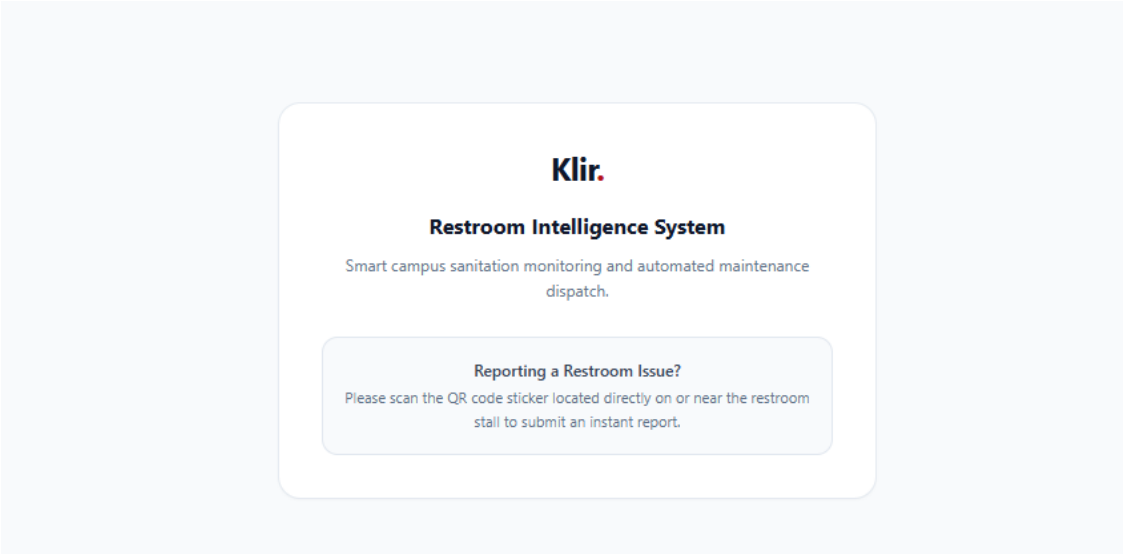
The user interface was designed as two connected applications: the web dashboard for administrative operations, and Klir Mobile for supervisor and technician field workflows. Both applications read from and write to the same Firestore records, so an action completed on one client is reflected on the other in real time.

Web Dashboard



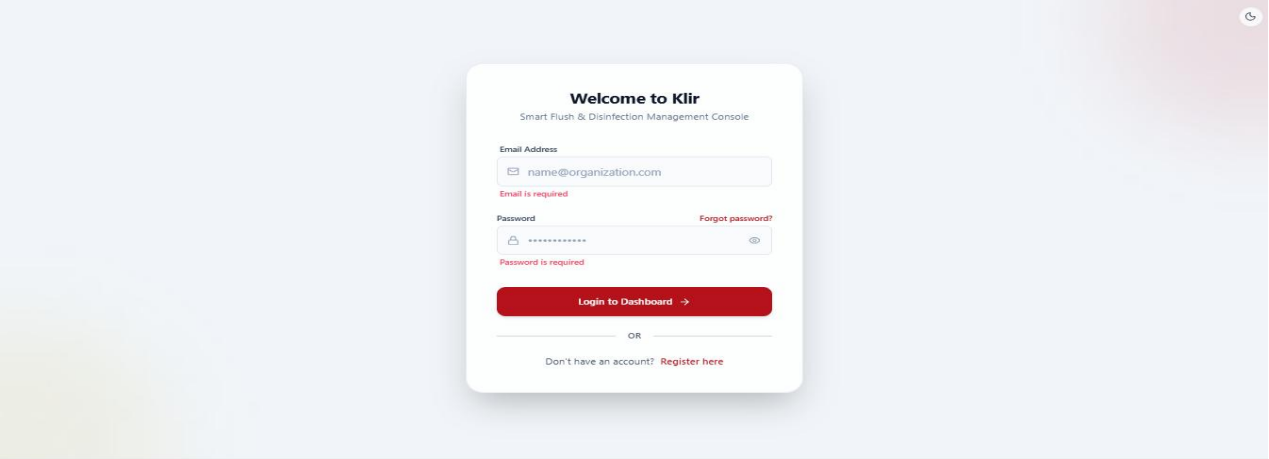
Public Issue Reporting Page

This page opens automatically when a student or faculty member scans the QR code sticker inside a restroom stall. It shows the specific restroom and stall the code belongs to, then lets the person pick the type of problem they noticed from a short list (for example, a broken lid, no water, a leak, or a UV-C light that isn't working), add an optional description, and optionally take a live photo with their phone's camera. No account or login is required. Once submitted, the report goes straight into the admin's Issue Reports queue, and the system temporarily blocks duplicate reports for the same stall while one is already pending.



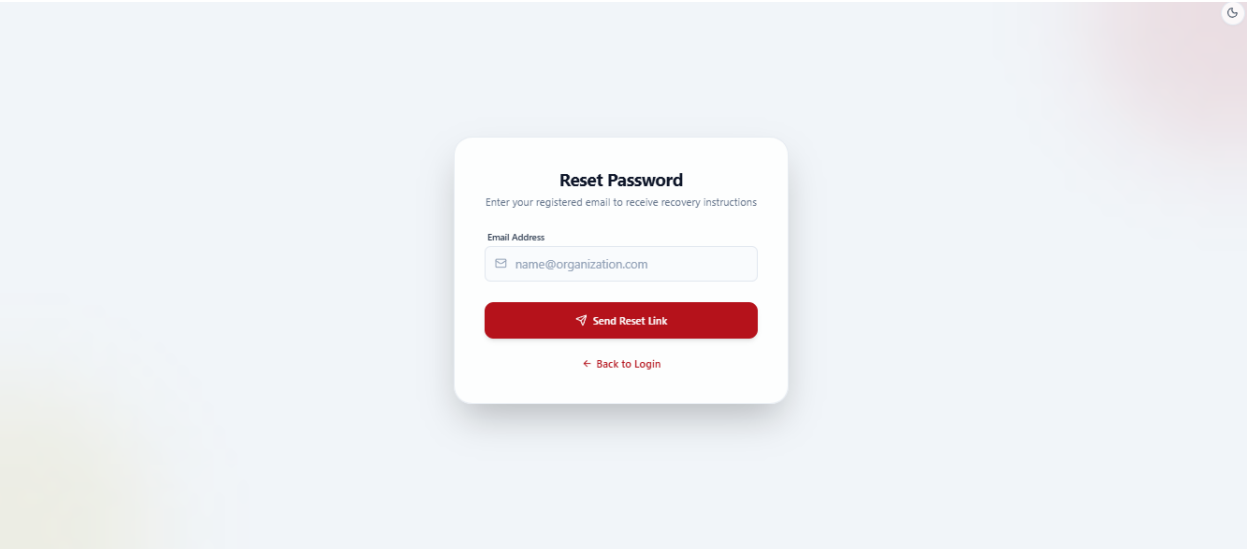
Root Landing Page

This is the public front door of the Klir web application. Visitors such as students and faculty see a short introduction to the touchless restroom system and are told to scan the QR code sticker inside a stall if they need to report a problem. If the browser already has an active staff session, the page skips the introduction and sends the user straight to the Operations Dashboard.



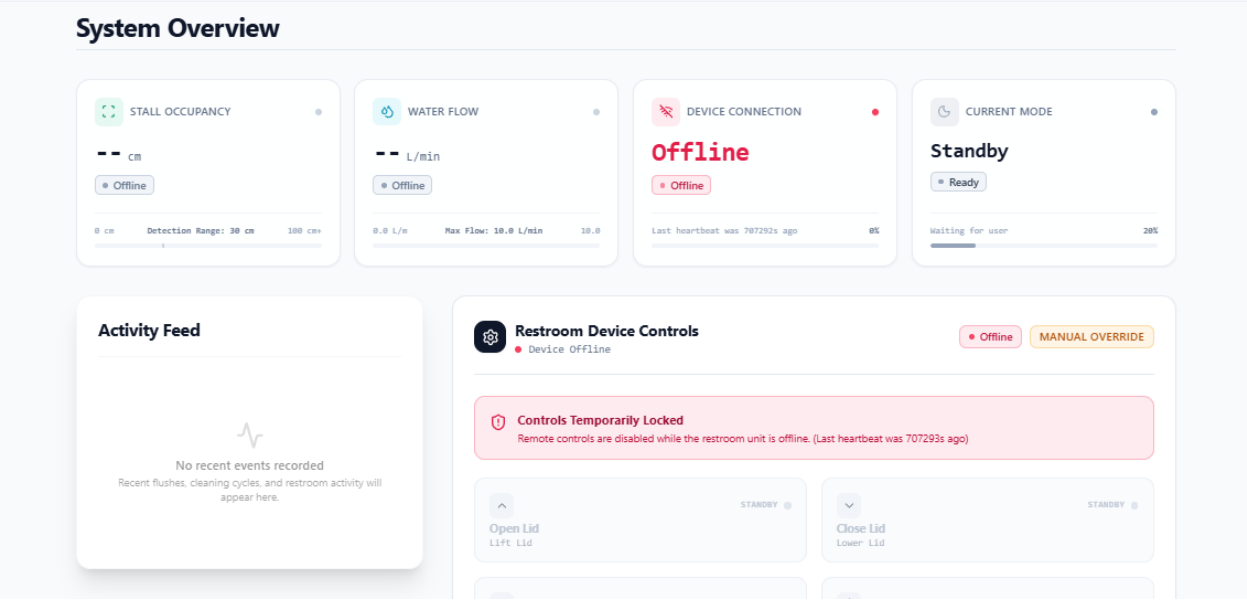
Admin Login Page

This is the entry point for campus administrators, the only role authorized to access the web dashboard. It asks for an email and password, checks the credentials against Firebase, and confirms the account holds an admin role before granting entry. Supervisors and maintenance personnel do not use this login; their workflows are handled entirely through Klir Mobile. Repeated failed login attempts are automatically rate-limited to prevent brute-force attacks.



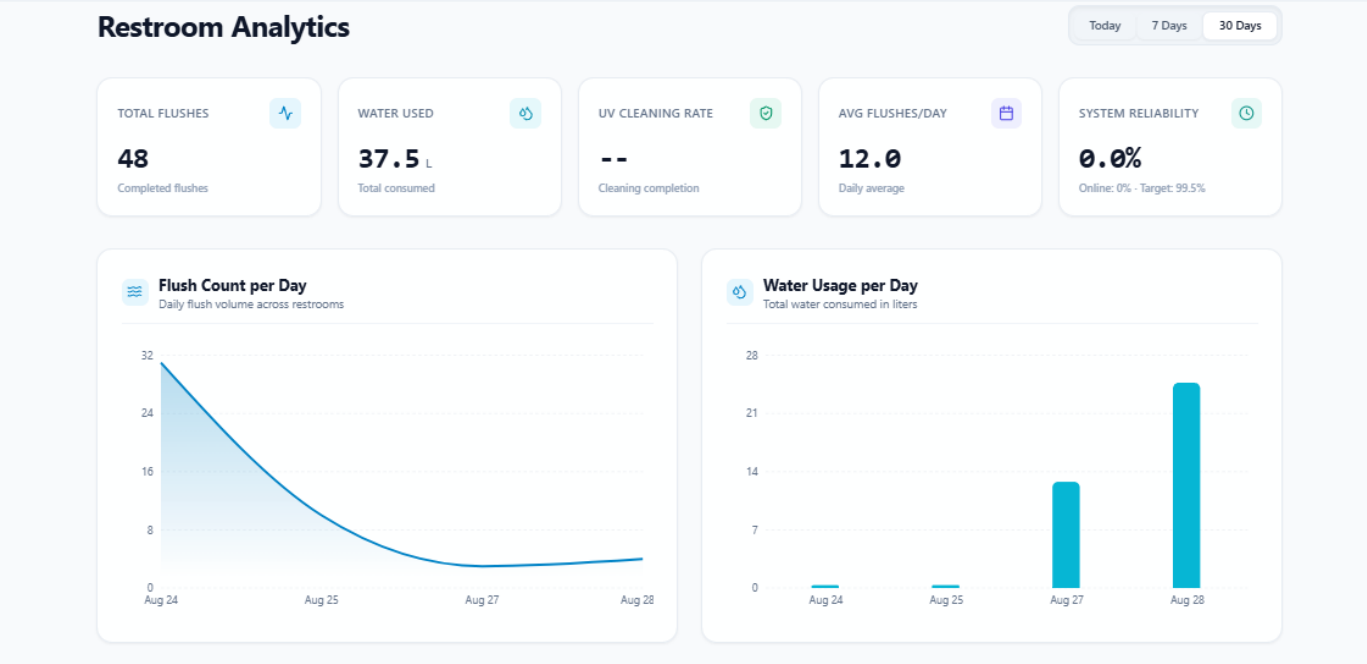
Reset Password Page

This page opens from the link in the password reset email. The link carries a one-time code that the page checks before showing the form, so only the person who received the email can use it. The user types a new password, confirms it, and is returned to the login page once the change is saved.



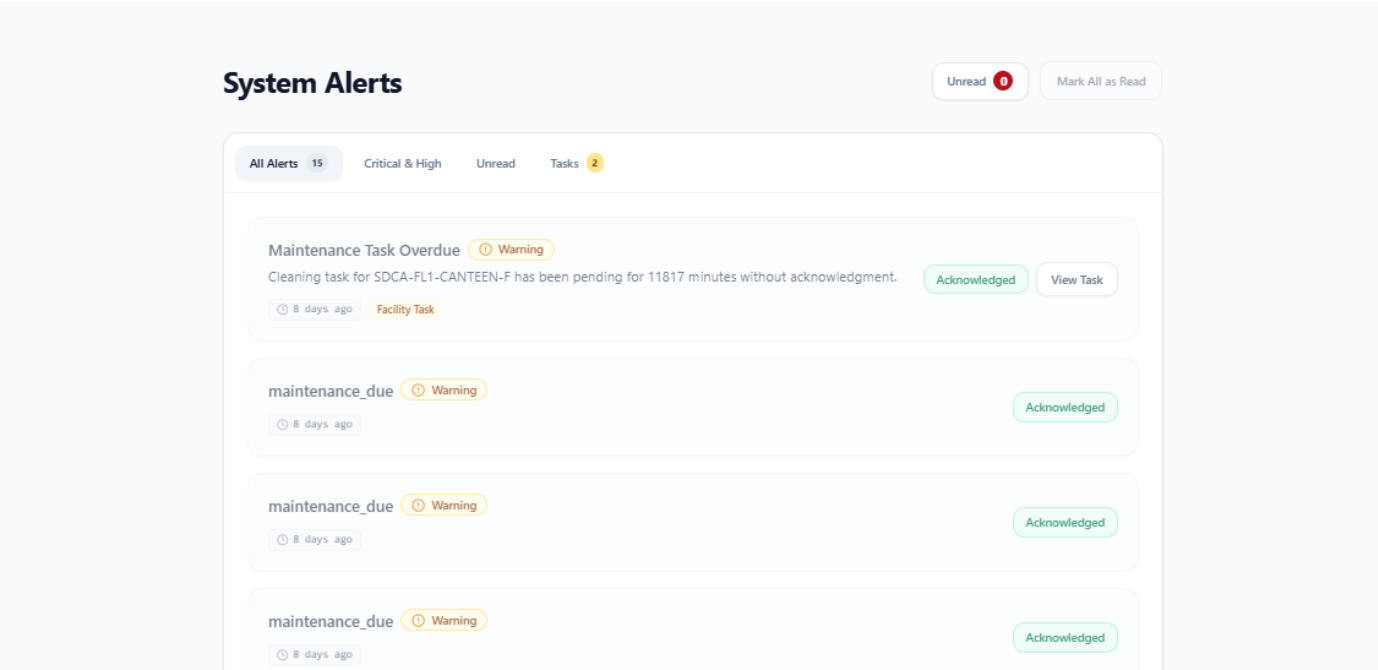
Operations Dashboard

This is the main command center for administrators. At the top, a set of summary cards shows the day's total flushes, how much water has been conserved, how many UV-C disinfection cycles have completed, and how many restroom devices are currently online versus offline. The dashboard also includes manual controls for flushing, running the UV-C light, and opening or closing the lid remotely, each protected by a confirmation step so nothing fires by accident. A live activity feed at the bottom shows a running log of sensor events, actuator activity, and maintenance updates as they happen.



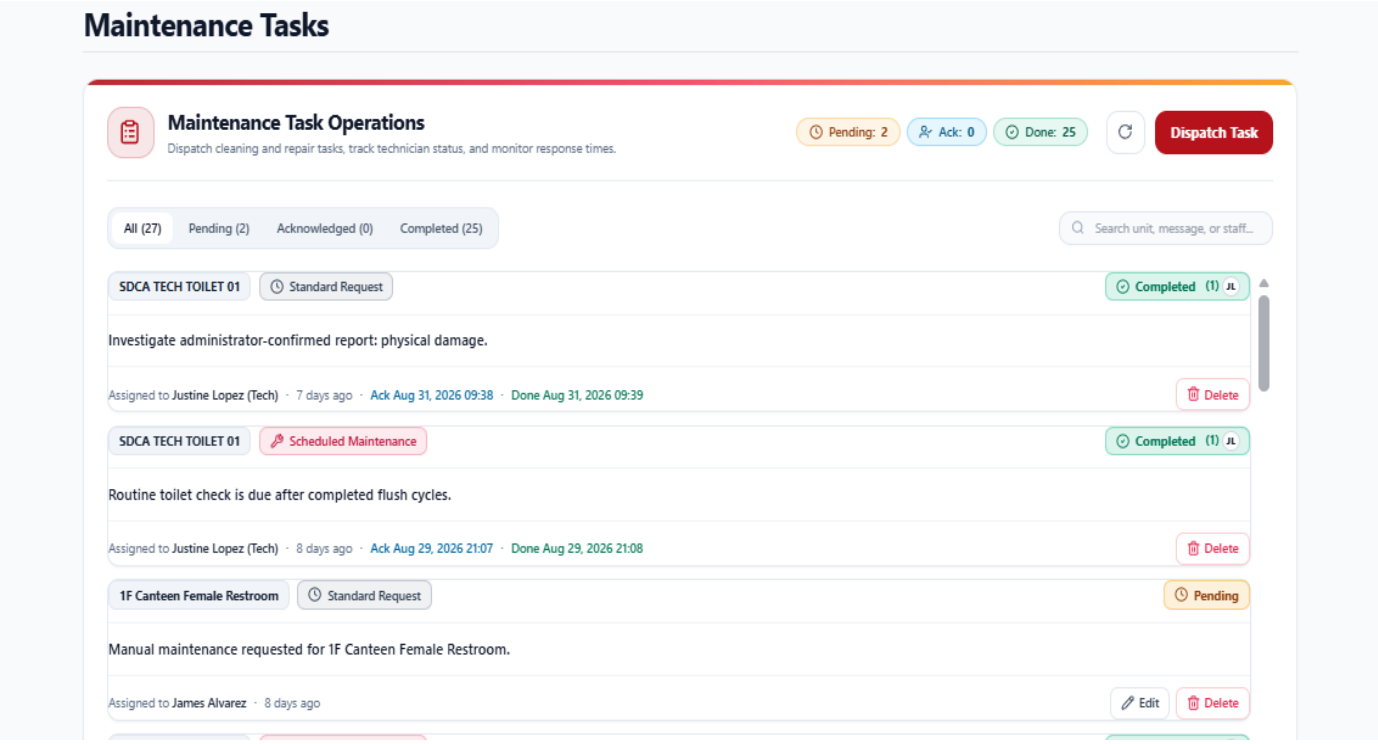
Analytics Page

The analytics page is where administrators dig into usage trends over time. It shows water consumption charts, a breakdown of which hours of the day see the most restroom traffic, and a comparison of flush activity across the four SDCA Annex floors. Date filters let staff switch between today, the last 7 days, the last 30 days, or a custom date range to spot patterns and plan cleaning schedules accordingly.



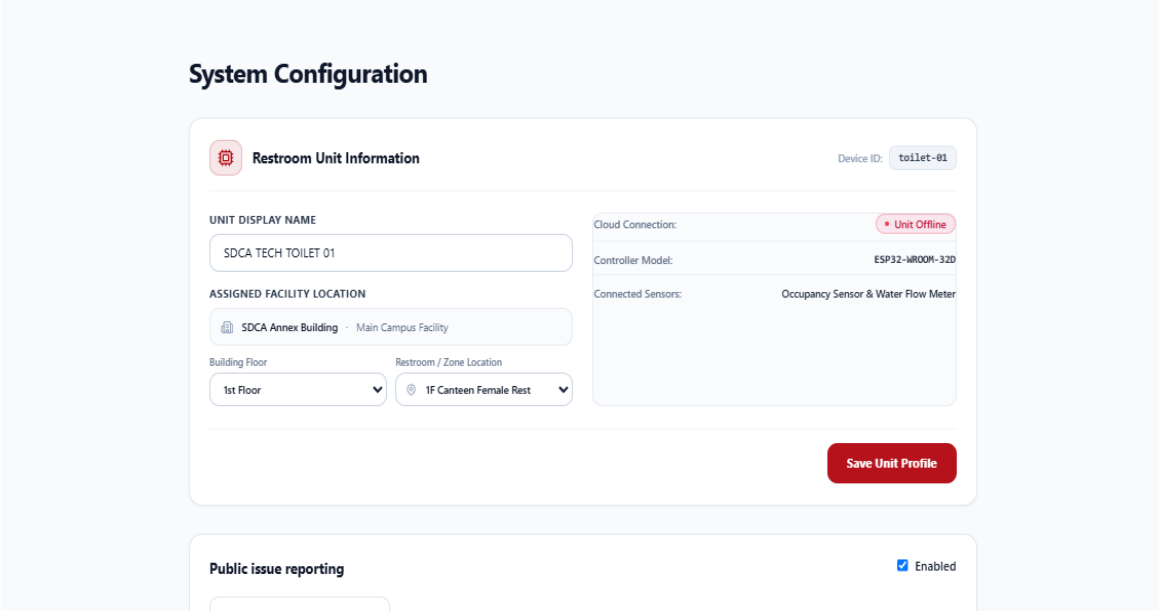
Alerts Page

This page centralizes every automated warning the system generates, things like a sensor going offline, water running continuously longer than it should, or a maintenance task that's gone unacknowledged for too long. Alerts are grouped by severity (critical, warning, or informational) so the most urgent issues stand out first, and staff can acknowledge one alert at a time or clear the whole list with a single click.



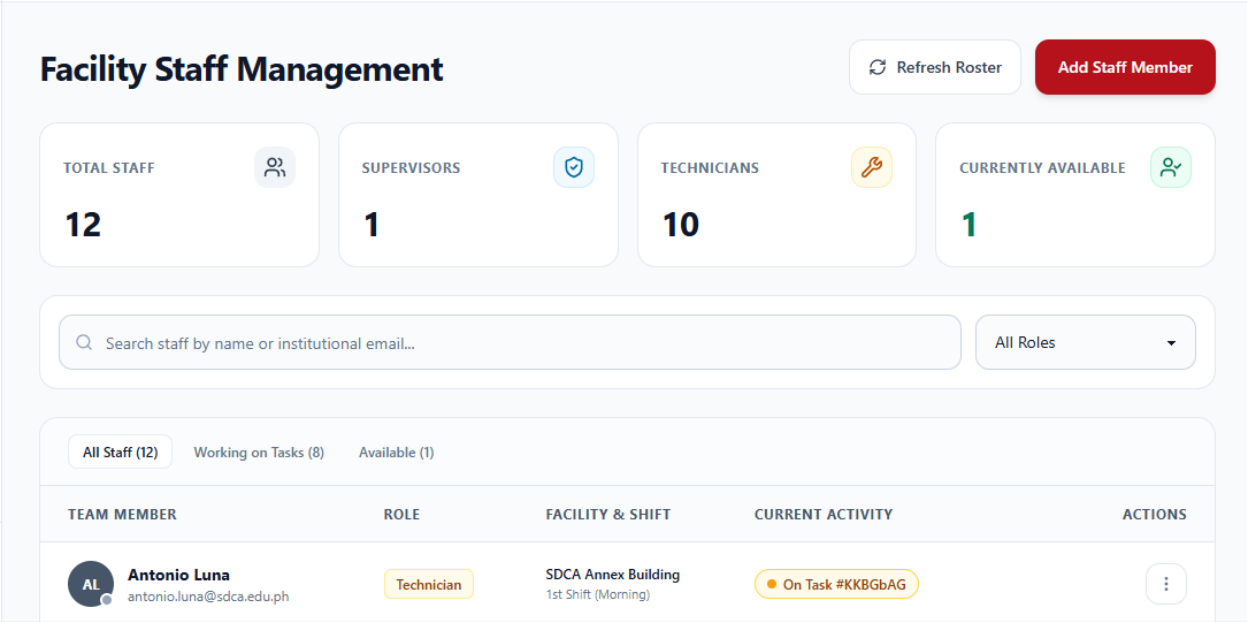
Tasks Page

The Tasks page is where all custodial work orders live, from the moment they're created to the moment they're closed out. Supervisors can see every task's current stage waiting for a technician, in progress, completed and awaiting review, or flagged for a recheck and can dispatch new tasks or reassign existing ones directly from this screen. Technicians attach before and after photos to the tasks they complete. Supervisors review these photos and either approve the task or flag it for a recheck with a note explaining what needs to be redone. Tasks can be created automatically by the system, for example after a UV-C failure or when a flush threshold is reached, or manually by a supervisor.



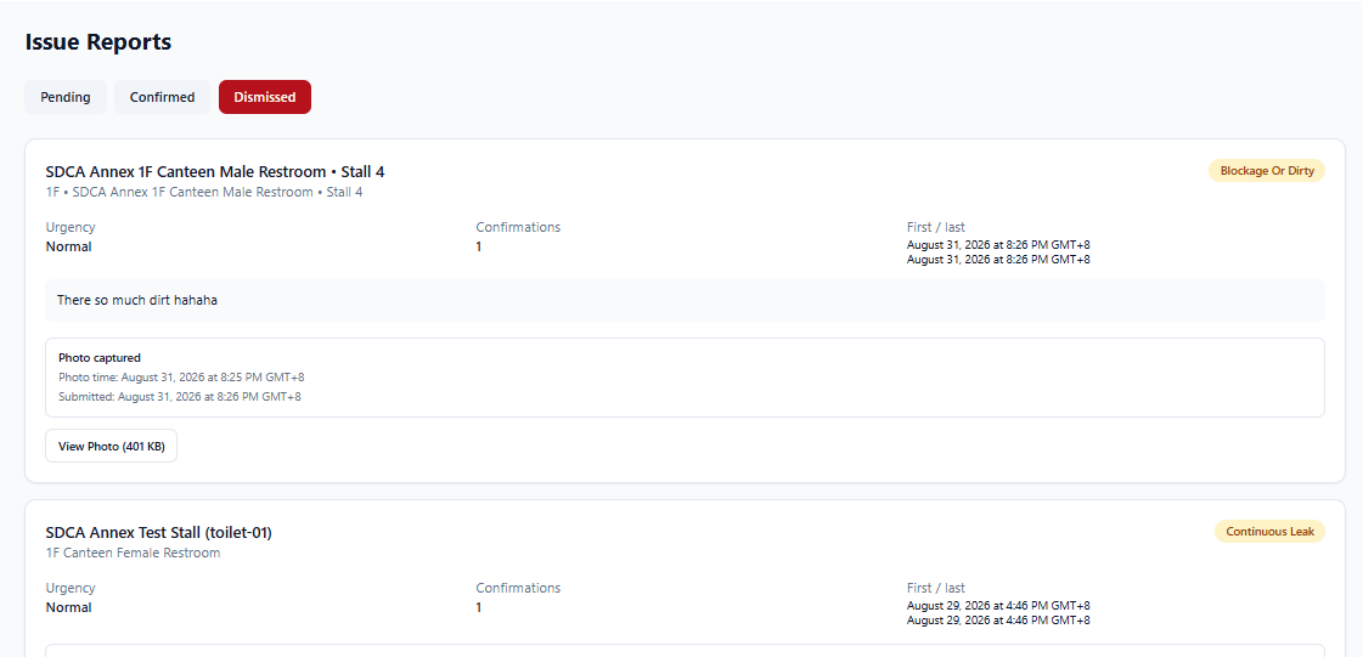
Configuration Page

This page is reserved for administrators and controls the more technical side of the system. It lets staff adjust hardware timing values (like how long the flush or UV-C cycle runs), set up automated rules (like automatically creating a cleaning task after a certain number of flushes, or flagging a leak), and generate printable QR code stickers for all 96 stalls in one click. It also has a toggle to turn public QR reporting on or off for any individual device.



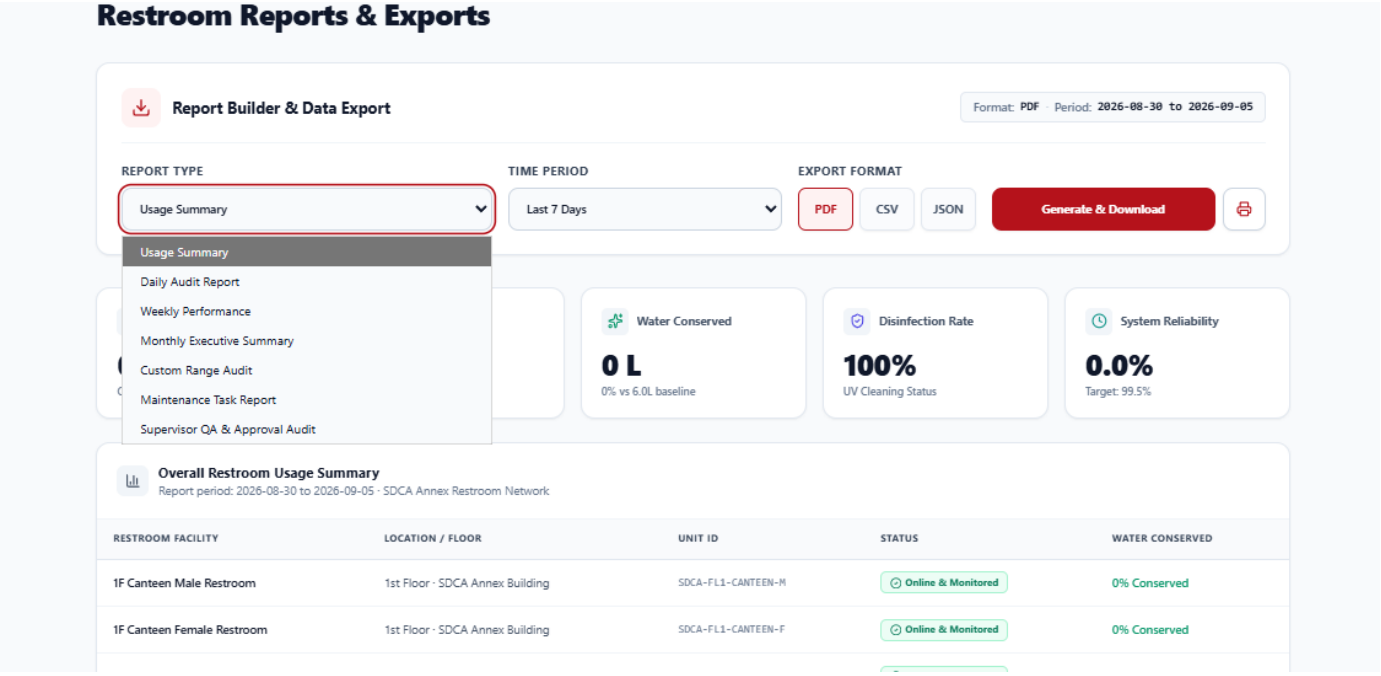
Staff & Personnel Management Page

This page is reserved for administrators and is where staff accounts are created and managed. It lists every maintenance technician, supervisor, and administrator, and for technicians it shows whether they are currently available and which work orders they are handling. Administrators can add new personnel by assigning them to a facility (for example, the SDCA Annex Building) and a shift (1st Morning, 2nd Evening, 3rd Night, or Rotational). They can also edit a person's details and facility coverage, and deactivate or reactivate an account. Changes to accounts require the administrator to re-enter their password.



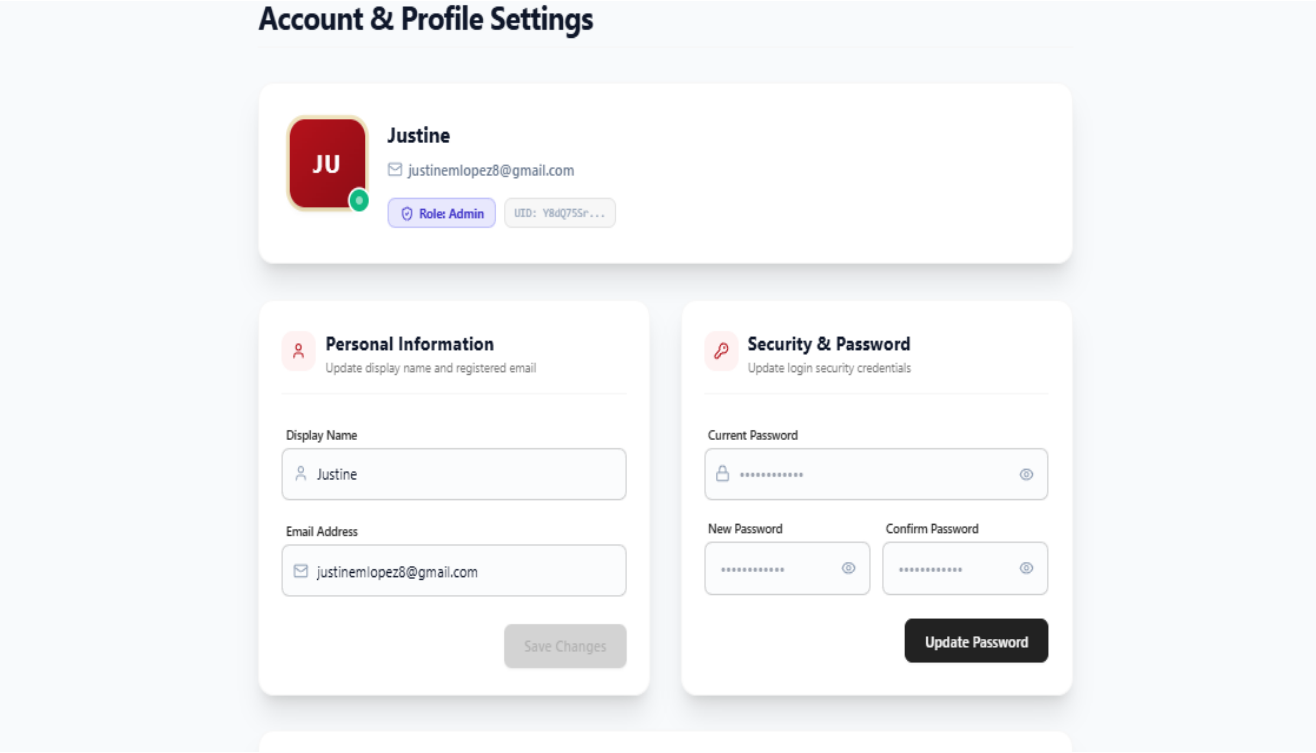
Issue Reports Page

This is where administrators review problems reported by students and faculty through the QR code stickers. Reports are sorted into three tabs Pending, Confirmed, and Dismissed and each report shows the photo evidence and description submitted by the reporter. Clicking "Confirm" on a report automatically turns it into a maintenance task assigned to that restroom, so the issue moves straight into the custodial workflow without extra manual entry. Clicking "Dismiss" closes out false alarms or duplicate reports and frees up that stall so new reports can be submitted again.



Reports & Exports Page

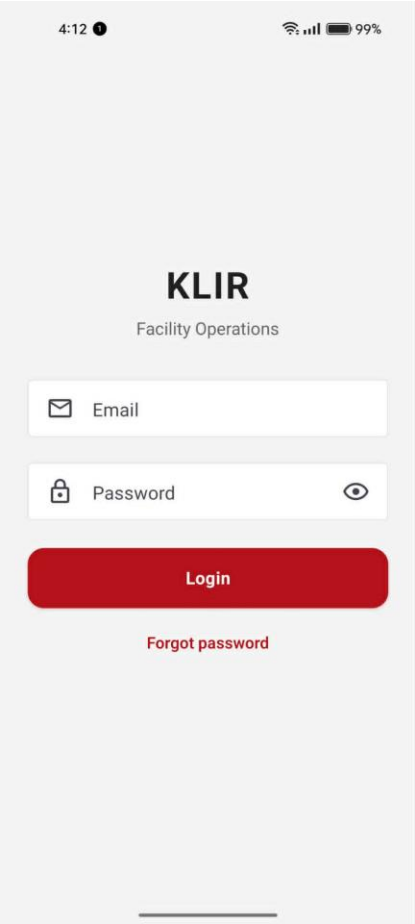
This page lets staff generate and download official reports in PDF with the SDCA letterhead, CSV, or JSON format. Four report types are available: Daily Restroom Activity, which gives hour-by-hour flush volume and telemetry; Restroom Usage and Water, which summarizes water saved, flush counts, and UV-C cycles; Cleaning and Maintenance Tasks, which covers technician turnaround times and completions; and Supervisor Audit, which covers inspections, approvals, and recheck rates. Every generated report is kept in an export history list so it can be downloaded again with one click.



Profile Page

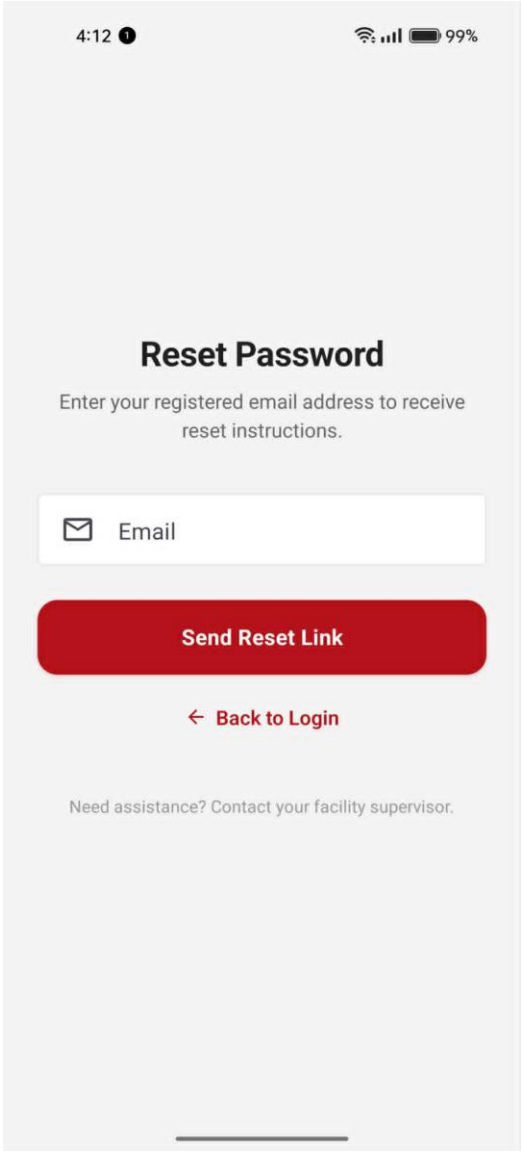
Every logged-in user has access to this page to view their account details, including their name, email, and assigned role. From here, users can also update their password and switch between light and dark themes

Klir Mobile



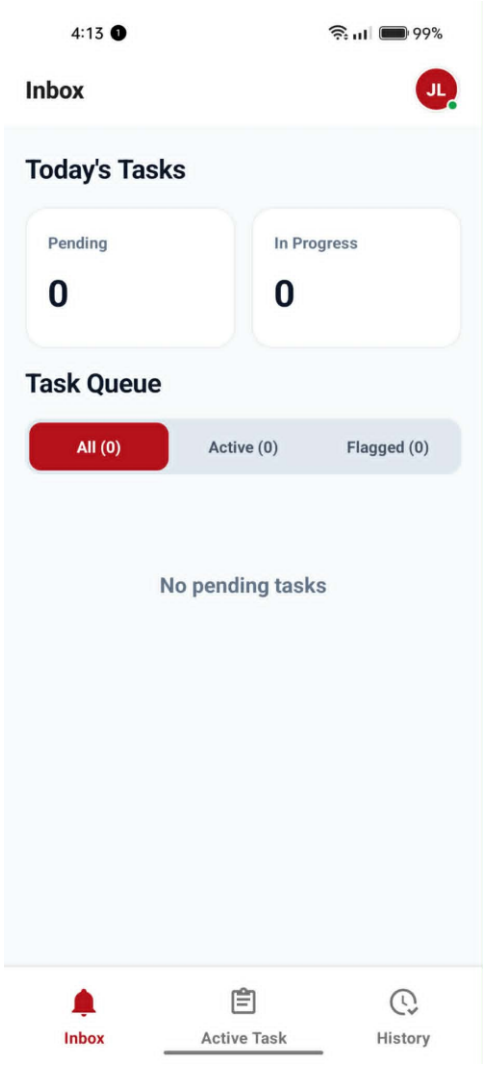
Login Screen

This is the first screen technicians and supervisors see when opening the app. It asks for an email and password, and also offers a one-tap biometric login (fingerprint or face unlock) if the user has previously signed in on that device. Once verified, the app automatically sends the user to either the technician workspace or the supervisor hub based on their role.



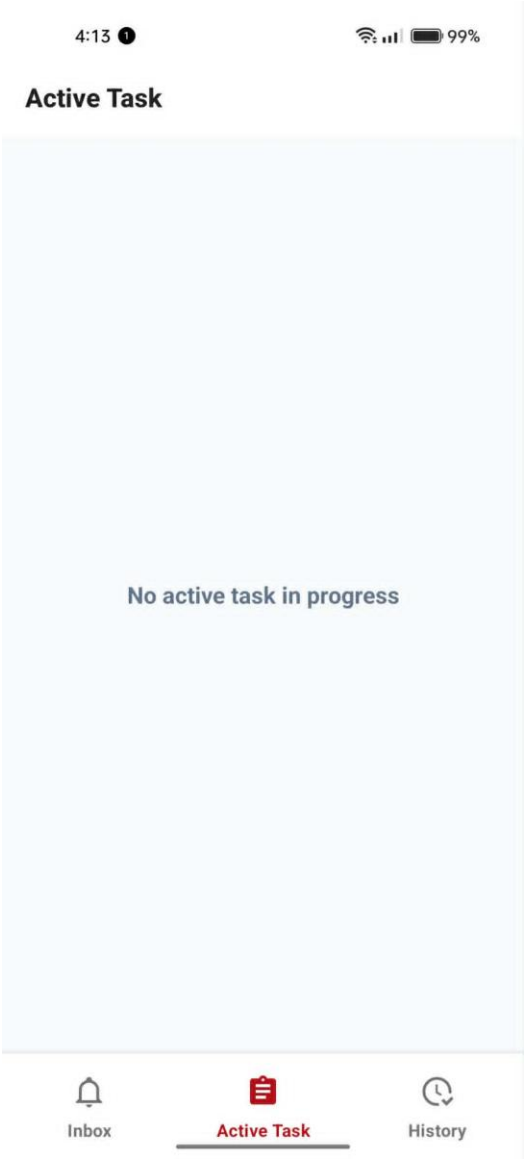
Forgot Password Screen

If a technician or supervisor forgets their password, they can request a reset link from this screen by entering their registered email.



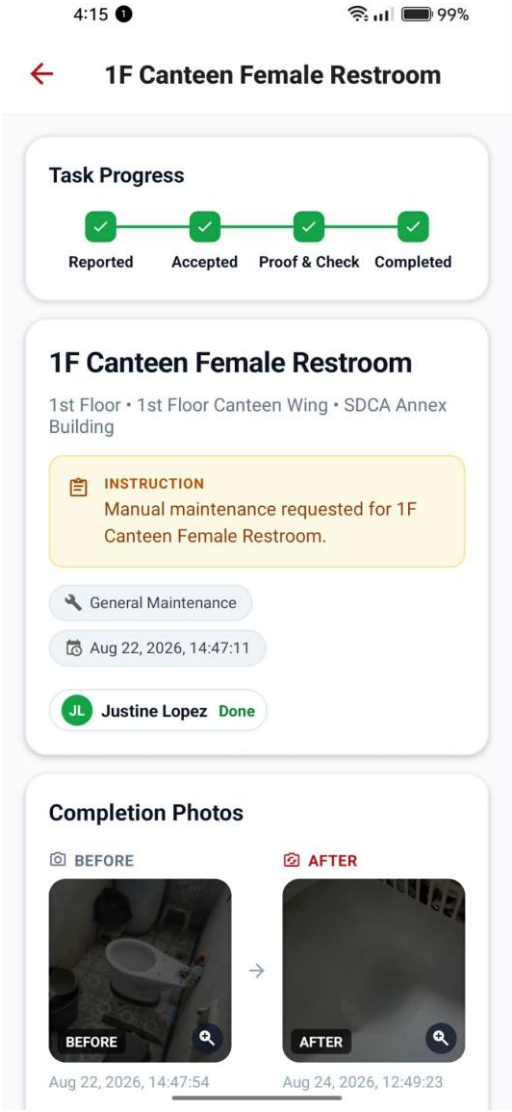
Inbox Screen

This is the technician's main hub for receiving work. It lists incoming tasks under three tabs All, Active, and Flagged so a technician can quickly see everything assigned to them, what they're currently working on, and anything a supervisor sent back for a recheck. If a hardware failure is detected somewhere on their floor (like a stuck valve or a broken pump), a bright red priority alert appears at the top of the inbox so it can't be missed.



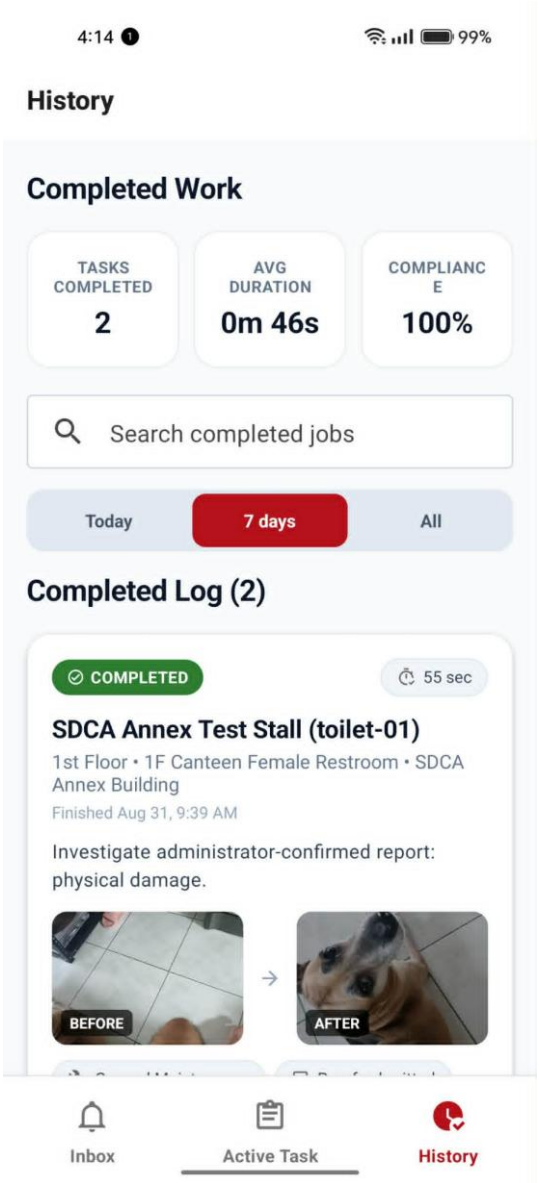
Active Task Screen

This screen shows the one task a technician is currently working on, without the distraction of the rest of their inbox. It displays the restroom location, a description of exactly what needs to be done, and a button to resume the cleaning process where they left off.



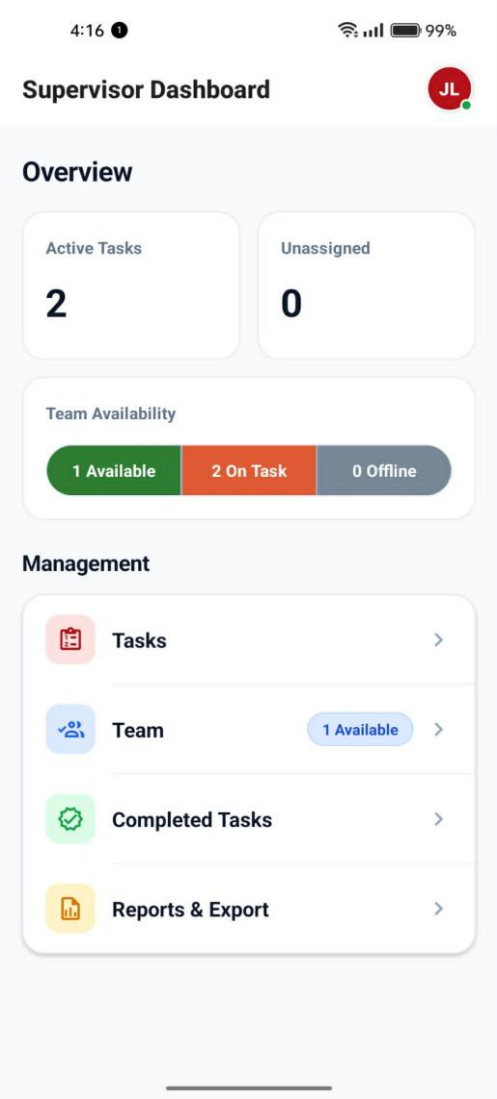
Task Detail Screen

This is the screen where a technician actually completes their work. It walks them through taking an arrival photo, checking off each item on the 10-point cleaning checklist, and then taking a final photo once the cleaning is done. A quick "Check All" button lets them mark everything as complete at once if needed, though everything can still be reviewed and adjusted before submitting.



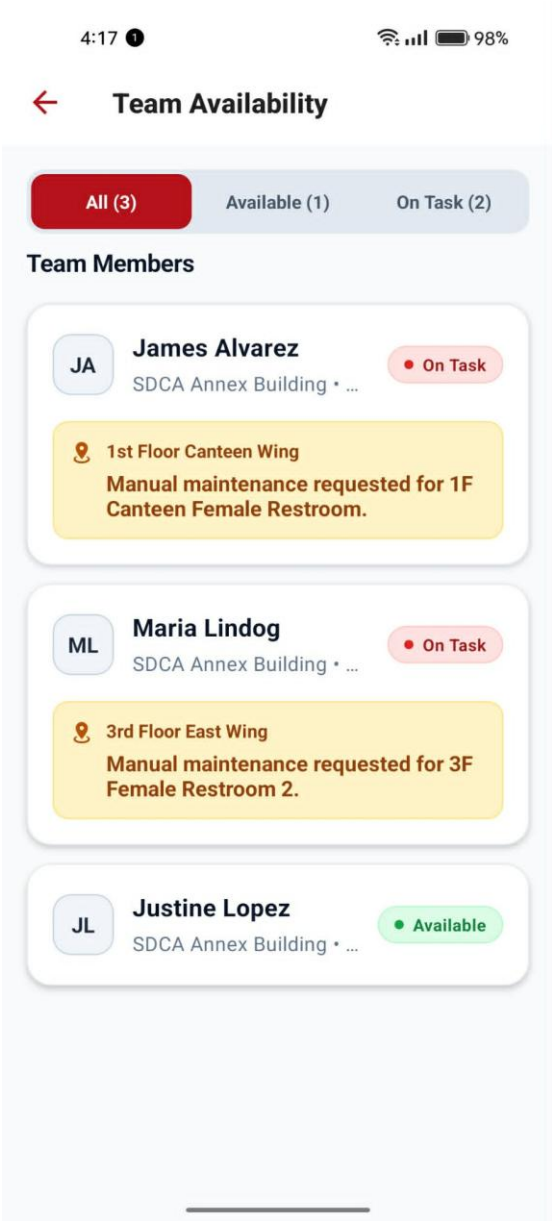
History Screen

Technicians can look back at everything they've completed on this screen. It shows how many tasks they've finished, their average completion time, and a scrollable list of past tasks with before-and-after photos attached to each one. A search bar and simple date filters (today, last 7 days, or all time) make it easy to find a specific past task.



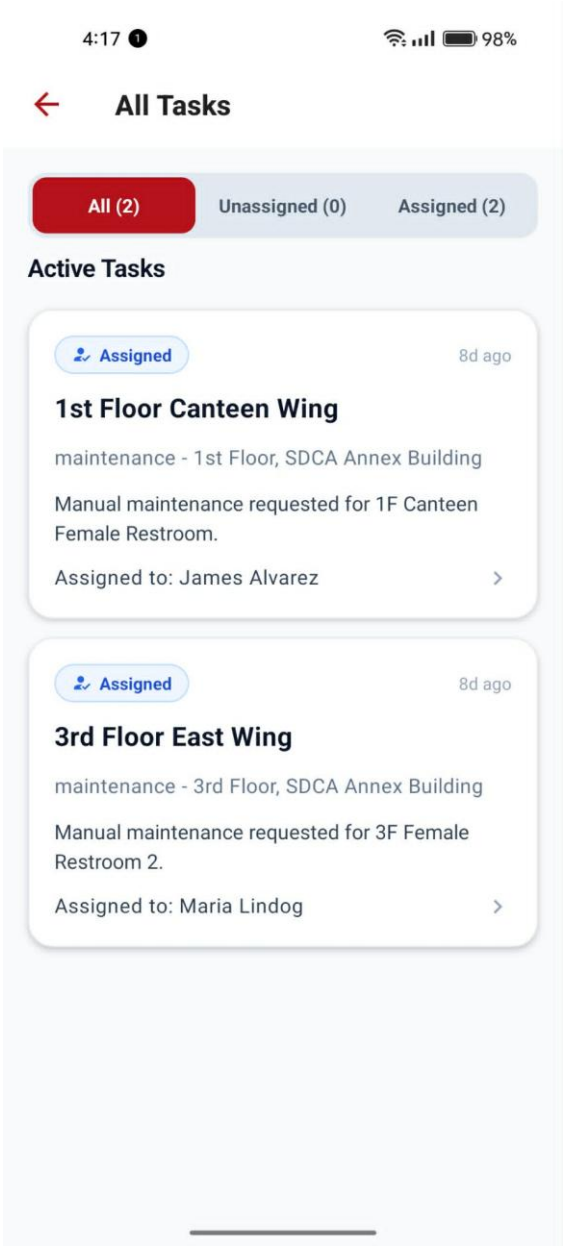
Supervisor Dashboard Screen

This is the main hub for supervisors. It shows a quick overview of how many tasks are currently active and how many still need a technician assigned, along with a live snapshot of how many staff members are available, currently working, or offline. From here, supervisors can jump into the team roster, the full task board, completed work waiting for review, or the reporting tools.



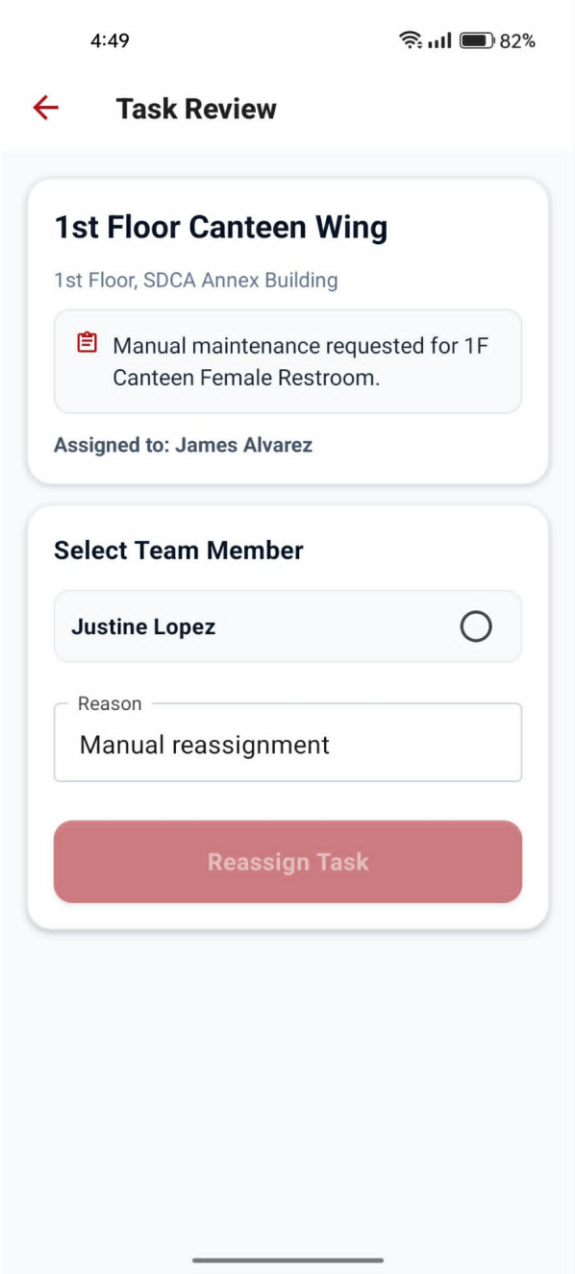
Team Availability Screen

This screen gives supervisors a live view of every technician's current status available, on task, or offline along with what restroom they're currently working in and how long they've been there. Supervisors can filter the list and reach out to a technician directly by phone or email from this screen.



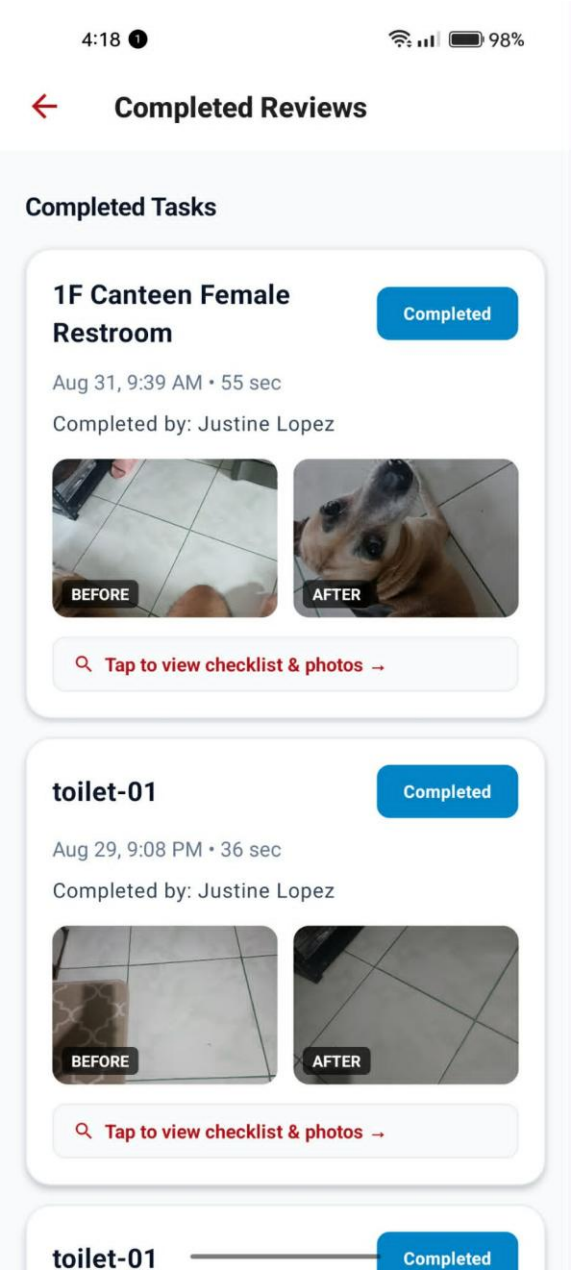
Supervisor Tasks Screen

This is the full list of every work order happening across the campus, viewable by supervisors. Tasks can be filtered by status (unassigned, in progress, or flagged), and each card shows the restroom, floor, and assigned technician at a glance.



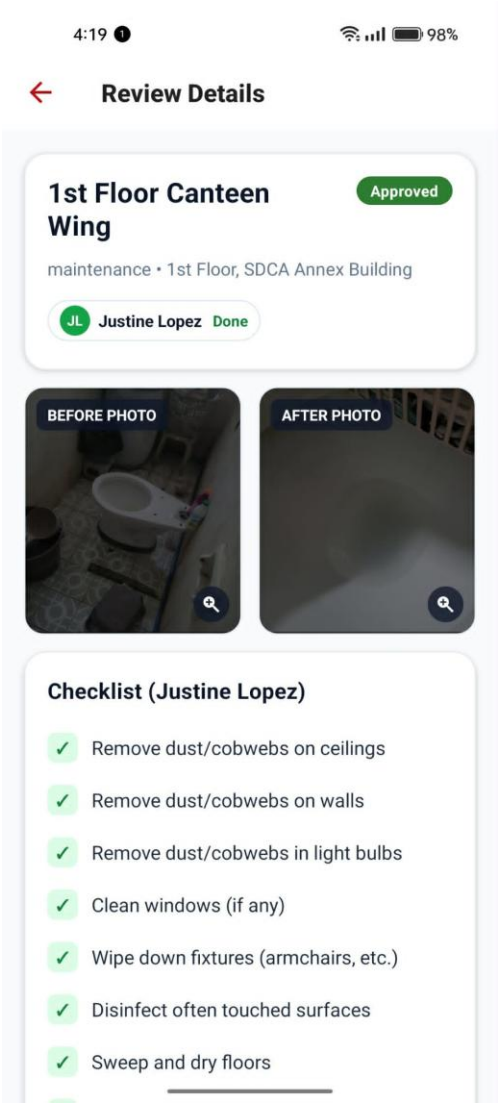
Supervisor Task Detail Screen

Tapping into a specific task from the tasks board opens this screen, which gives supervisors a closer look at that individual work order including sensor readings that may have triggered it and how long it's been sitting unaddressed. From here, a supervisor can reassign the task to a different technician or flag it for urgent attention.



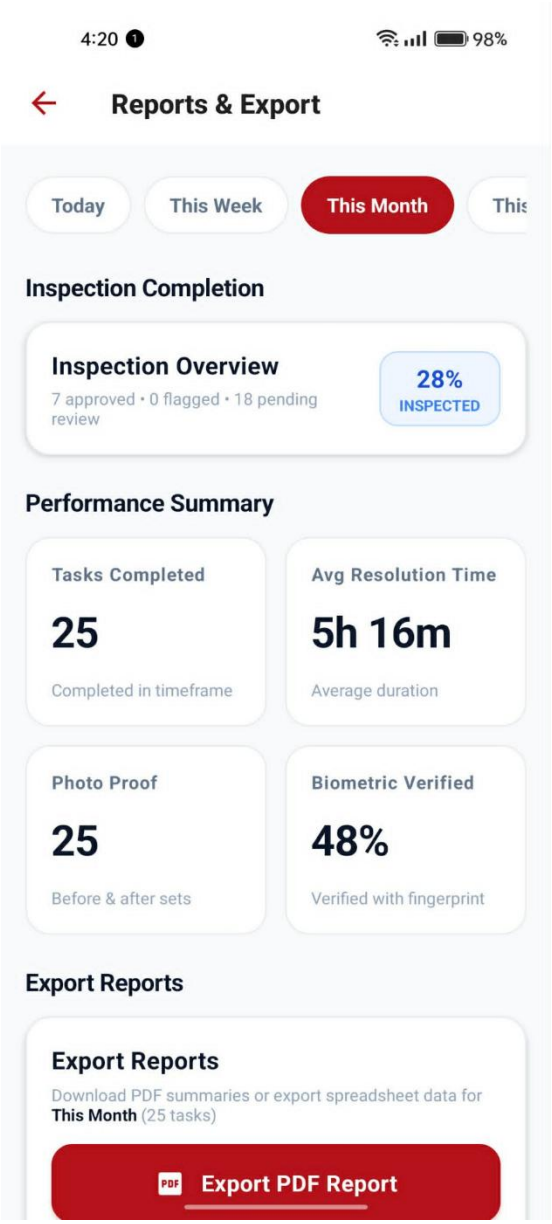
Completed Reviews Screen

This screen holds every finished task that's waiting for a supervisor's final sign-off. Because this involves reviewing sensitive work records, the app requires a fingerprint, face scan, or password before allowing access. Tasks are grouped into pending review, approved, and flagged for recheck.



Completed Review Detail Screen

This is where a supervisor actually inspects a technician's finished work. It shows the before-and-after photos side by side, along with any extra photos the technician took of specific areas like stalls or sinks, and a full readout of the 10-point checklist confirming what was completed. From here, the supervisor either approves the task and closes it out, or flags it and sends it back to the technician with a note on what still needs fixing.



Supervisor Reports Screen

This screen lets supervisors generate the same kind of compliance reports available on the web dashboard, but from their phone. It shows a summary of completion rates, average cleaning time, and how quickly technicians responded to alerts, and lets the supervisor export everything as a PDF or CSV file to share.

a.4.2. System Input and Output

Input	Source	Result / Output
Ultrasonic distance	ESP32 sensor	Occupancy status and state-machine transition
Water-flow pulses	YF-S201 sensor	Volume/flow event and possible continuous-flow alert
Manual actuator request	Authorized web user	MQTT command to lid, flush, UV-C, reset, or configuration topic
Cleaning dispatch	Admin / supervisor	Firestore work order and FCM notification
Task acknowledgment	Technician mobile app	Updated task status and personnel availability
Checklist and photo proof	Technician mobile app	Submission record and uploaded evidence
QA decision	Admin / supervisor	Approved status or flagged recheck with remarks
Automation rule	Admin configuration	Alert or work-order generation when a condition is met
Report request	Web dashboard	PDF, CSV, or JSON output
Heartbeat / last seen	ESP32 / listener	Online or offline device indication

a.4.3. Role-Based Access Control and Security

Capability	Admin	Supervisor	Maintenance	Viewer
Web dashboard	Full	No	No	No
Mobile app	No	Command hub	Workspace	No

Create / dispatch tasks	Yes	Yes	No	No
Approve / flag QA	Yes	Yes	No	No
Reassign tasks	Yes	Yes	No	No
Complete tasks	No	Audit only	Yes	No
Hardware overrides	Full	No	No	No
Automation rules	Edit	View	No	No
Export reports	Yes	Yes	No	View only

The role matrix is enforced through authenticated user records, Firebase custom claims where applicable, and server-side verification of bearer tokens on protected API routes. Critical hardware and configuration actions are not exposed to maintenance or viewer accounts. The mobile application further protects the completed-review area with biometric authentication when available and a password fallback when biometrics cannot be used.

a.4.1. REST API and Data Services

The current web backend exposes 39 REST endpoints organized by task management, supervisor actions, devices, maintenance notes, actuator commands, sensors, analytics, reports, alerts, automation rules, and authentication. The large number of endpoints reflects the separation of responsibilities in the final system and allows the frontend to apply role-specific permissions to each operation.

Group	Representative Operations
Tasks	List, create, read, update, delete, acknowledge, accept recheck, complete, register notification token

Supervisor	Approve task, flag task, reassign task
Devices	List devices, register device, get device details, get connection status
Actuators	Flush, UV-C, lid open, lid close, reset
Sensors / Analytics	Readings, statistics, configuration, dashboard summary, flush patterns, water usage, system performance
Reports	Generate and download reports
Alerts / Rules	List and acknowledge alerts; create, update, delete, and reset automation rules
Authentication	Login, register, logout, current user, password reset

B. PROJECT DEVELOPMENT

b.1. Gantt Chart and Development Sequence

The project timeline covers the major phases from February to August 2026. Planning and requirements definitions were conducted from February 9 to February 28, followed by system design from February 15 to March 4. Development began on March 1 and continued through August 24 as the team integrated and refined the hardware, web, cloud, and mobile components. Testing ran in parallel from March 13 through August 24, while the initial deployment was completed on March 30 to 31. Maintenance activities continued from March 31 through August 24. A mobile application redeployment was scheduled for August 24 to 25 to apply the latest mobile updates. Development, testing, and maintenance are marked with an asterisk because these activities were still ongoing at the time of documentation.

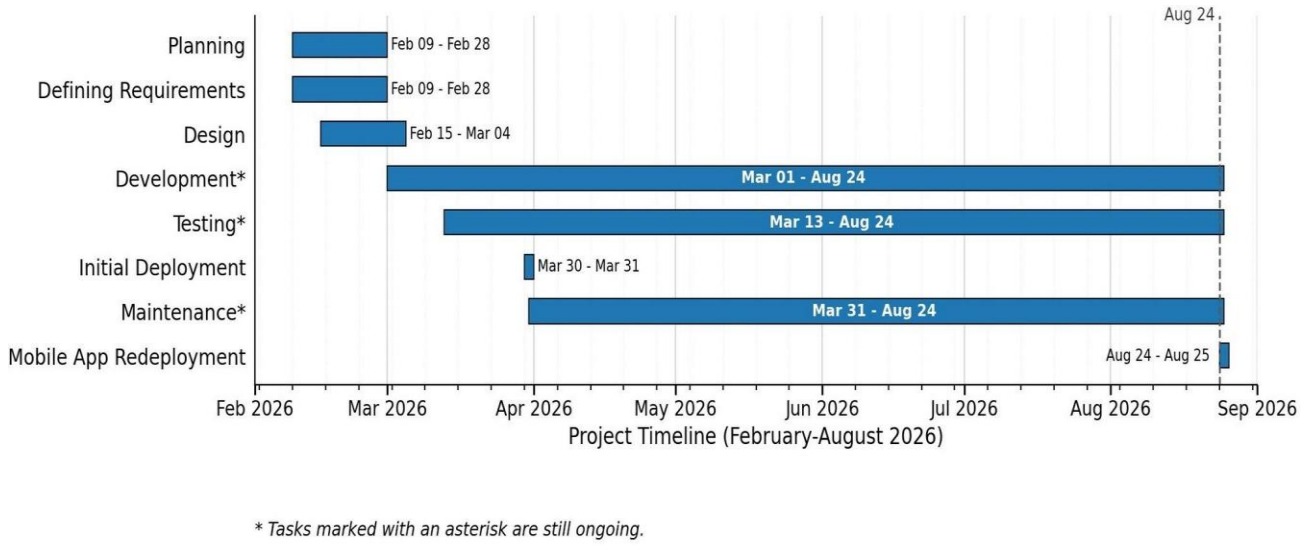


Figure 7. Gantt chart of the KLIR project development timeline from February to August 2026. Tasks marked with an asterisk are still ongoing.

b.2. Testing Process

Testing was performed at the web, mobile, cloud, and hardware levels. The current project no longer relies only on manual inspection. Automated Jest tests are part of both software codebases, while integration checks are still needed for functions that depend on a physical ESP32, Android device, MQTT broker, or cloud deployment. For the real UV-C hardware, the project also reviewed the supplied HP350 spectral irradiance reports as supporting component-level verification. These reports are treated separately from the project team's functional testing because they document optical output under their own stated test conditions.

The web project reports 67 passing Jest unit and integration tests. The mobile project reports 149 passing tests across 19 Jest test suites. These automated tests are used to verify application logic and reduce regression when task management, authentication,

supervisor functions, and technician workflows are changed. Additional project scripts are available for TypeScript checking, production builds, linting, Playwright end-to-end testing, security auditing, and Expo diagnostics. The presence of these commands does not by itself replace live hardware and mobile-device verification, so both automated and functional testing remain part of the validation process.

b.2.1. Web Dashboard and Backend Testing

Area	Validation
Authentication and RBAC	Verify valid login, invalid access, role restrictions, protected API routes, and re-authentication for critical functions.
Task management	Verify creation, dispatch, update, reassignment, acknowledgment state, completion state, and recheck handling.
Supervisor QA	Verify approve and flag actions, audit filters, approval-rate calculation, and supervisor remarks.
Device telemetry	Verify Firestore updates, latest readings, device last-seen state, and offline timeout behavior.
Automation and alerts	Verify usage-count rules, continuous-flow alerts, alert acknowledgment, and hardware counter updates.
Actuator commands	Verify confirmation dialogs and MQTT command publication for flush, lid, UV-C, configuration, and reset.
Reporting	Verify generation of supported report types and PDF/CSV/JSON exports.
Automated tests	Run the 67-test Jest suite and confirm all tests pass.

b.2.2. Mobile Application Testing

Area	Validation
Role routing	Supervisor account opens the command hub; maintenance account opens the technician workspace.
Task inbox	Verify task filtering, deduplication, cached hydration, active/flagged states, and empty states.
Reassignment	Verify that only unacknowledged tasks can be reassigned and that technician availability is updated.
Three-step execution	Verify arrival proof, 10-item checklist, completion proof, and submission validation.
Photo handling	Verify 1080 px resizing, 75% JPEG quality, overlays, additional area photos, and supervisor gallery viewing.
Biometric security	Verify biometric gate and password fallback for supervisor reviews; validate biometric login behavior where supported.
Offline queue	Verify queued acknowledgments/submissions and synchronization after connectivity returns.
Multi-assignee	Verify separate per-user submissions and task completion rules.
Automated tests	Run all 19 Jest suites and confirm 149 tests pass.

b.2.3. Hardware and Cloud Integration Verification

1. ESP32 connects to the configured Wi-Fi network and HiveMQ Cloud TLS endpoint.
2. Ultrasonic telemetry is published and received by the MQTT listener.

3. Water-flow events are recorded in Firestore with the expected unit and volume fields. 4 Lid, flush, and UV-C events are written as device events and appear in the dashboard activity feed.
4. Authorized commands from the web application reach the correct MQTT command topic.
5. UV-C manual activation is blocked when the occupancy condition is unsafe.
6. The listener creates alerts or work orders when configured rules are triggered.
7. FCM notification delivery is confirmed on a registered physical Android device.
8. Device heartbeat/last-seen behavior marks a silent unit offline after the configured timeout.
9. Task changes made on the mobile application appear on the web dashboard through the shared Firestore records.

UV-C spectral documentation was reviewed in addition to the functional interlock test. The HP350 reports documented peak wavelengths of 258.3 nm and 276.4 nm and UV-C irradiance values of 736.88 µW/cmZ and 174.77µW/cmZ at 30 mm for the two RZXLED modules. This was accepted as evidence of UV-C source output, but not as evidence that the five-second KLIR cycle delivers a validated germicidal dose to all toilet surfaces.

b.2.4. Unit and Integration Testing

Unit tests focus on isolated application logic such as authentication helpers, task-state functions, data validation, custom hooks, filters, role checks, and utility functions. External services such as Firebase and MQTT are mocked when a test needs to isolate the local behavior. Integration tests cover combinations of

components such as the task UI with Firestore data, supervisor screens with task state, and technician execution flows with submission validation.

b.2.5. End-to-End and Acceptance Testing

End-to-end validation checks the complete flow from a physical event or administrative action to the final visible result. For example, a sensor event should travel from the ESP32 to HiveMQ Cloud, through the listener, into Firestore, and then appear on the dashboard. A dispatched cleaning task should be written to Firestore, sent through FCM, displayed on the technician device, completed with evidence, and then become available for supervisor review.

Acceptance Area	Current Criterion
Core web automated tests	67 Jest tests passing
Core mobile automated tests	149 tests passing across 19 suites
Role separation	Admin/supervisor/maintenance/viewer permissions enforced for documented functions
Safety-sensitive UV-C control	Occupancy protection and confirmation controls enabled
Task QA loop	Dispatch -> acknowledge -> execute -> submit -> approve/flag supported
Multi-location support	22 SDCA Annex restrooms registered plus one test unit
Cloud deployment model	Vercel web + Railway listener + Firebase + HiveMQ Cloud
Accessibility	Web interface designed to WCAG 2.2 AA requirements documented by the project
UV-C spectral verification	Supplied HP350 reports document UV-C

spectral output at 258.3 nm and 276.4 nm; no claim of in-situ germicidal dose or microbial reduction.

b.3. Installation and Deployment

The production architecture separates the web application from the MQTT ingestion service. The Next.js dashboard and API routes are deployed to Vercel, while the MQTT listener runs continuously on Railway. HiveMQ Cloud provides the TLS MQTT broker, and Firebase provides authentication, Firestore, Storage, and Cloud Messaging. This arrangement allows the system to keep a persistent device connection without depending on a serverless process that may stop after a request ends.

The mobile application is built using the Expo development and Android build workflow. It connects to the same backend API and Firebase project used by the web application. Native notification and biometric functions should be tested using a physical Android device because emulator behavior does not fully represent FCM token delivery and device-level authentication.

b.3.1. Cloud Deployment Requirements

Node.js 20 or 22 LTS and npm 10 or later for the web project.

Firebase project with Authentication and Firestore enabled; Firebase Storage and FCM are required for the full mobile proof and notification workflow.

HiveMQ Cloud cluster configured for TLS on port 8883.

Railway service for the long-running MQTT listener.

Vercel project for the Next.js dashboard and API routes.

Android Studio / Android SDK and an Expo-compatible Android device for native mobile validation.

b.3.2. Firebase Project Configuration

Firebase client configuration is provided to the web and mobile applications through environment variables. Server-side Firebase Admin credentials are kept in protected server environments and must not be embedded in client code or committed to the source repository. Firestore collections store devices, sensor readings, tasks, alerts, automation rules, maintenance notes, and users. Storage is used for mobile proof images, while FCM tokens are registered for supported supervisor and maintenance accounts.

b.3.3. HiveMQ Cloud MQTT Configuration

The MQTT broker uses TLS port 8883 with authenticated credentials. The ESP32 publishes sensor and event topics, and the Railway listener subscribes to the required topic patterns. The web backend publishes authorized actuator commands to the command topics. Topic names must remain synchronized across firmware, listener code, and server API logic to avoid silent communication failure.

b.3.4. Backend Service Deployment

The mqtt-listener directory is deployed as a separate Railway service. Its production process maintains the MQTT connection, handles graceful shutdown, evaluates alerts, writes Firestore batches, updates hardware counters, and dispatches FCM notifications. Environment variables include the broker host, username, password, port, device identifier, and Firebase Admin credentials.

b.3.5. Web Dashboard Deployment

The Next.js application is deployed to Vercel. The deployment includes the protected dashboard routes and the REST API endpoints used by the web and

mobile clients. After deployment, the team verifies authentication, Firestore access, task management, report generation, actuator command publication, responsive behavior, and role restrictions before considering the release stable.

b.3.6. ESP32 Firmware Deployment

Firmware is uploaded to the ESP32 through the Arduino/PlatformIO development environment. Before connection to the full hardware assembly, the team verifies GPIO assignments, external power, common ground, ultrasonic echo voltage reduction, relay behavior, and the UV-C occupancy interlock. After upload, serial output and the dashboard are used together to confirm Wi-Fi connection, MQTT connection, telemetry publication, state-machine transitions, and that the UV-C relay can only energize under the intended unoccupied condition. The supplied HP350 reports are retained as spectral documentation for the UV-C source, while any final operational deployment should separately verify irradiance at representative toilet surfaces.

b.3.7. Mobile Application Deployment

Klir Mobile is installed through the Expo/Android build process for controlled institutional testing. The application requires the correct API base URL, Firebase configuration, and Google services file. After installation, the team verifies role-based login, FCM token registration, task synchronization, image capture and upload, biometric behavior, offline queue behavior, and supervisor review.

b.3.8. Cloud Deployment Workflow

1. Configure Firebase, HiveMQ Cloud, Vercel, Railway, and protected environment

variables.

2. Deploy the MQTT listener to Railway and verify that the service remains connected to the broker.
3. Deploy the Next.js dashboard to Vercel and verify authentication and API routes.
4. Upload the current ESP32 firmware and confirm telemetry publication.
5. Build and install the Android mobile client and register notification tokens.
6. Run the end-to-end workflow from device event to dashboard and from work-order dispatch to supervisor QA review.
7. Keep the previous stable deployments available for rollback if a new release fails.

b.3.9. Rollout Phases

Phase	Activity
1	Cloud configuration and security setup
2	MQTT listener deployment and broker validation
3	Web dashboard deployment and API validation
4	ESP32 installation and device registration
5	Mobile supervisor/technician client setup
6	Integrated testing for each restroom/device group
7	Controlled operational use, log review, and issue correction

b.3.10. Rollback and Recovery Plan

Component	Recovery Method

Web dashboard	Revert to a previous Vercel deployment and restore the last known working environment configuration.
MQTT listener	Redeploy or restart the last stable Railway build after reviewing service logs.
Mobile application	Reinstall the last stable Android build on controlled devices.
ESP32 firmware	Re-upload the last stable firmware and restore the known MQTT/configuration values.
Firestore	Use exported records or available backup procedures for important production data; separate test data from operational data.
Credentials	Rotate compromised MQTT, Firebase, or application credentials and invalidate affected sessions.

b.4. Maintenance

Maintenance covers the physical hardware, cloud services, web and mobile software, data records, credentials, and operational rules. Because the system crosses several technologies, a maintenance change in one part must be followed by a short integration check to confirm that the connected modules still agree on topic names, data fields, task states, and permissions.

b.4.1. Warranty Period

For academic deployment, the project team should define a support period during which implementation defects found in the demonstrated system are corrected without changing the agreed capstone scope. Physical component damage caused by water, misuse, incorrect power, or unauthorized modification should be assessed separately from software defects.

b.4.2. Cloud Service Monitoring

Vercel, Railway, Firebase, and HiveMQ Cloud should be checked regularly for deployment errors, quota warnings, authentication problems, service interruption, and unusual message activity. Railway logs are especially important because the listener is responsible for the persistent MQTT connection, alert evaluation, Firestore writes, and FCM dispatch.

b.4.3. Web Dashboard Maintenance

Dashboard maintenance includes checking authentication, the 22-unit facility directory, telemetry cards, alerts, QA audit filters, reports, automation rules, and hardware controls. After a dependency or UI update, the 67-test Jest suite should be run together with the production build and any relevant browser or end-to-end checks.

b.4.4. Backend Service Maintenance

The listener and API services should be reviewed for MQTT errors, Firebase errors, invalid payloads, FCM failures, and task-state conflicts. Dependency updates should be tested before production deployment, and environment variables should be validated whenever a service is moved or rebuilt.

b.4.5. MQTT Communication Maintenance

MQTT maintenance includes verifying broker credentials, TLS connectivity, publish/subscribe topics, message payload format, reconnect behavior, and device heartbeat activity. A topic change must be coordinated across the ESP32, listener, and server command code.

b.4.6. Firebase Firestore and Storage Maintenance

Firestore growth, invalid test records, duplicate tasks, security rules, index needs, and read/write usage should be reviewed. Mobile proof images stored in Firebase Storage should follow a retention plan because image evidence can grow more quickly than text records.

b.4.7. User Account and Security Maintenance

Administrators should review active users and roles, remove unused accounts, rotate compromised credentials, protect service-account keys, and verify that the role matrix still prevents maintenance or viewer accounts from performing supervisor or actuator functions. Biometric access does not replace Firebase authentication; it acts as an additional local gate for supported mobile functions.

b.4.8. Mobile Application Maintenance

Mobile maintenance includes running the 149-test suite, TypeScript checks, Expo diagnostics, notification tests on a physical device, biometric tests, image upload tests, offline queue tests, and task synchronization checks.

Updated builds should be validated with both supervisor and maintenance accounts before distribution.

b.4.9. ESP32 Firmware Maintenance

Firmware maintenance includes checking Wi-Fi and MQTT reconnection, sensor calibration, state-machine timing, flow measurement, actuator output, command handling, and last-seen messages. The code should be re-tested after changes to thresholds, GPIO assignments, message schemas, or power hardware.

b.4.10. Hardware Component Maintenance

The physical installation should be inspected for moisture, loose wiring, power instability, relay wear, sensor obstruction, valve condition, servo resistance, UV-C light degradation, and enclosure damage. High-current loads must remain isolated from the ESP32 logic supply, and the ultrasonic echo voltage protection should be checked after rewiring. For long-term UV-C use, the institution should also verify that the light source remains securely enclosed and consider periodic irradiance checks because UV output can change with operating time, contamination, component replacement, and installation geometry.

b.4.11. Bug Classification and Response

Critical issues include complete system outage, uncontrolled actuator behavior, unsafe UV-C activation, authentication failure affecting all users, or corrupted task state. High-priority issues include failed notifications, missing sensor records, supervisor QA failures, or repeated task duplication. Medium and low issues include report errors, visual defects, and non-blocking usability problems. Each issue should include reproduction steps, logs or screenshots, severity, action taken, and resolution status.

b.4.12. Preventive Maintenance Schedule

Daily checks should confirm that telemetry and task data are still arriving. Weekly checks should include MQTT connectivity, push notifications, role login, supervisor review, technician submission, and hardware inspection. Monthly checks should review Firestore/Storage usage, user accounts, dependency updates, and backup/export needs. A full end-to-end validation should be performed after

major updates and before a formal demonstration or operational rollout.

b.4.13. Data Retention and Backup

Recent sensor data should be kept for analytics, while old test records should be archived or removed so they do not distort reports. Completed tasks and supervisor audit results should be retained longer because they form the maintenance history. Source code should remain backed up through version control, and important Firestore data should be exported according to the capabilities of the Firebase project.

b.4.14. Long-Term Support and Limitations

A larger institutional deployment may require paid cloud plans, stronger environmental enclosures, power backup, formal UV-C electrical and exposure-safety certification, in-situ irradiance and dose validation, plumbing engineering, centralized mobile distribution, and microbiological testing if the institution wants to make verified disinfection claims. The supplied HP350 reports establish that the selected light modules produce UV-C output under their stated test conditions, but they do not by themselves certify the complete KLIR installation as a germicidal device. The current capstone demonstrates the technical integration and operational workflow, while these additional validation requirements should be completed before treating the system as a fully certified public-restroom product.

CHAPTER 4

PRESENTATION OF DATA, ANALYSIS AND INTERPRETATION

This chapter presents, analyzes, and interprets the data gathered from the respondents who evaluated Klir: An IoT-Based Toilet Sanitation, Analytics, and Predictive Maintenance Management System. The evaluation used the ISO/IEC 25010 software quality model, covering Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The data are presented in tabular form and arranged according to the eight quality characteristics. Each characteristic has five paired statements, one from the End-User questionnaire and one from the Expert questionnaire.

4.1 Profile of the Respondents

The study gathered responses from 100 respondents: 20 IT experts (20%) and 80 end-users (80%). In the questionnaire data, end-users were recorded under the combined evaluator category "Student/Maintenance Team," while experts were recorded under "Faculty/IT Expert." Separate counts by student and janitor role, interface used, and expert specialization were not collected in the administered form.

Table 1. Distribution of Respondents

Respondent Group	Frequency	Percentage
IT Experts	20	20%
End-Users	80	80%
Total	100	100%

4.2 Data Gathering Procedure

The two questionnaires were administered separately. Each questionnaire has 40 statements (8 characteristics with 5 statements each) rated on a five-point Likert scale.

The IT experts reviewed the technical documentation and the recorded technical demonstration of the system, which covered the ESP32 hardware prototype, MQTT communication, Firestore backend, web dashboard, and mobile application. They then answered the Expert Evaluation Questionnaire.

The end-users, recorded in the questionnaire as Student/Maintenance Team, installed the Klir mobile application and performed guided tasks using individual test accounts. The tasks were logging in and navigating the application, acknowledging a work order, completing the checklist and proof photos, using the application in offline mode, and viewing task history. They then answered the End-User Evaluation Questionnaire.

Both questionnaires were administered through Google Forms. Respondents were informed that their responses would be used only for academic purposes.

4.3 Statistical Treatment of Data

The weighted mean was computed per statement, per characteristic, and overall using the formula $\bar{x} = \Sigma x / N$.

The standard deviation was used to determine the consistency of the responses using the formula $SD = \sqrt{[\Sigma(x - \bar{x})^2 / (N - 1)]}$.

The Pearson correlation coefficient (r) was computed between the End-User and Expert characteristic means to determine whether the two groups' perceptions are aligned, using the formula $r = [n\Sigma xy - (\Sigma x)(\Sigma y)] / \sqrt{\{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]\}}$.

Cronbach's alpha (α) was used to test the internal consistency of each characteristic and of

each questionnaire, using the formula $\alpha = [k / (k - 1)] \times [1 - (\Sigma\sigma^2_i / \sigma^2)]$, where k is the number of items, σ^2_i is the variance of each item, and σ^2 is the variance of the total score.

Table 2. Likert Scale and Verbal Interpretation

Scale	Range	Verbal Interpretation
5	4.21 - 5.00	Strongly Agree
4	3.41 - 4.20	Agree
3	2.61 - 3.40	Neutral
2	1.81 - 2.60	Disagree
1	1.00 - 1.80	Strongly Disagree

Table 3. Interpretation of Cronbach's Alpha

α Value	Internal Consistency
0.90 and above	Excellent
0.80 - 0.89	Good
0.70 - 0.79	Acceptable
0.60 - 0.69	Questionable
Below 0.60	Poor

Table 4. Interpretation of the Correlation Coefficient (r)

r Value	Strength
0.80 - 1.00	Very strong
0.60 - 0.79	Strong

r Value	Strength
0.40 - 0.59	Moderate
0.20 - 0.39	Weak
0.00 - 0.19	Very weak or negligible

4.4 Presentation and Analysis of Data

Each table below shows the End-User statement, its paired Expert statement, the computed mean, the standard deviation (SD), and the verbal interpretation (VI).

4.4.1 Functional Suitability

Table 5. Functional Suitability

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	Klir provides the occupancy detection, automated flushing, UV-C disinfection, and sanitation monitoring features I need to perform my tasks.	4.31	0.80	Strongly Agree
1	Expert	The system correctly implements occupancy detection, automated flushing, UV-C disinfection cycling, and sanitation analytics as specified in the project requirements.	4.60	0.60	Strongly Agree
2	End-User	The sensor readings, water-usage records, cleaning records, and sanitation status shown by Klir are accurate and reliable.	4.34	0.81	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
2	Expert	The system produces accurate sensor readings, water-usage records, cleaning records, and sanitation status based on captured inputs.	4.65	0.59	Strongly Agree
3	End-User	Klir performs tasks and shows options according to my assigned role and responsibilities.	4.36	0.77	Strongly Agree
3	Expert	Role-based business rules for Administrator, Supervisor, and Maintenance Personnel are properly implemented within the system.	4.55	0.69	Strongly Agree
4	End-User	The monitoring, alerts, cleaning tasks, reports, dashboard, and mobile features of Klir work together smoothly.	4.42	0.78	Strongly Agree
4	Expert	The ESP32 firmware, MQTT communication, Firestore backend, web dashboard, and mobile application function correctly as integrated modules.	4.60	0.68	Strongly Agree
5	End-User	Klir effectively helps me monitor restroom sanitation, respond to maintenance needs, and complete my assigned tasks.	4.35	0.84	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
5	Expert	The system consistently meets its intended functional objectives for automated toilet sanitation, monitoring, analytics, and maintenance support.	4.70	0.57	Strongly Agree

Category Mean - End-User: 4.36 (Strongly Agree); Expert: 4.62 (Strongly Agree). Analysis: End-users gave the Functional Suitability statements a mean of 4.36, while IT experts gave a mean of 4.62. For end-users, Pair 4 received the highest mean rating of 4.42, while Pair 1 received the lowest mean of 4.31. For experts, Pair 5 received the highest mean rating of 4.70, while Pair 3 received the lowest mean of 4.55. The expert mean was 0.26 points higher than the end-user mean.

Interpretation: Both groups interpreted the Functional Suitability characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.2 Performance Efficiency

Table 6. Performance Efficiency

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	The Klir dashboard or mobile application responds quickly when I use its features.	4.41	0.81	Strongly Agree
1	Expert	The web dashboard and mobile application respond within acceptable processing times under normal operating conditions.	4.60	0.68	Strongly Agree
2	End-User	Klir performs smoothly without noticeably slowing down the device I use.	4.34	0.84	Strongly Agree
2	Expert	System resources, including ESP32 processing, network communication, backend services, and database operations, are utilized efficiently during operation.	4.60	0.68	Strongly Agree
3	End-User	Klir remains responsive even when it is receiving live sensor updates and handling multiple activities.	4.39	0.77	Strongly Agree
3	Expert	The system maintains stable performance when multiple sensor readings, sanitation events, and user requests are processed concurrently.	4.70	0.57	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
4	End-User	Information, records, and reports in Klir load and update promptly when I need them.	4.34	0.87	Strongly Agree
4	Expert	Database operations are optimized for efficient storage and retrieval of sensor, water-usage, cleaning, and maintenance data.	4.70	0.57	Strongly Agree
5	End-User	Klir remains responsive even when a large amount of historical data is viewed.	4.36	0.78	Strongly Agree
5	Expert	The system supports increasing server traffic without performance degradation.	4.65	0.67	Strongly Agree

Category Mean - End-User: 4.37 (Strongly Agree); Expert: 4.65 (Strongly Agree). Analysis: End-users gave the Performance Efficiency statements a mean of 4.37, while IT experts gave a mean of 4.65. For end-users, Pair 1 received the highest mean rating of 4.41, while Pair 2 and Pair 4 received the lowest mean of 4.34. For experts, Pair 3 and Pair 4 received the highest mean rating of 4.70, while Pair 1 and Pair 2 received the lowest mean of 4.60. The expert mean was 0.28 points higher than the end-user mean.

Interpretation: Both groups interpreted the Performance Efficiency characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.3 Compatibility

Table 7. Compatibility

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	Klir shares up-to-date information correctly between the restroom device, web dashboard, and mobile application.	4.42	0.79	Strongly Agree
1	Expert	The system exchanges information effectively among the ESP32 device, MQTT broker, Firestore backend, analytics services, web dashboard, and mobile application.	4.60	0.68	Strongly Agree
2	End-User	Klir works properly with the supported browsers, mobile devices, and connected services required for my tasks.	4.44	0.78	Strongly Agree
2	Expert	The system adheres to established communication and integration standards and protocols, including MQTT over TLS and supported APIs.	4.65	0.67	Strongly Agree
3	End-User	Data shown in the Klir web dashboard and mobile application remains consistent, accurate, and complete.	4.39	0.88	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
3	Expert	Data interoperability is properly supported across the web dashboard, mobile application, backend, and analytics layer.	4.70	0.57	Strongly Agree
4	End-User	Using Klir does not interfere with the other applications or programs I use.	4.42	0.81	Strongly Agree
4	Expert	The system can coexist with other applications and normal network activity without operational conflicts.	4.60	0.68	Strongly Agree
5	End-User	Klir notifications and other connected features work as expected without unexpected errors.	4.33	0.92	Strongly Agree
5	Expert	External interfaces and connected services, including notifications, messaging, cloud database, and analytics functions, operate consistently and correctly.	4.70	0.57	Strongly Agree

Category Mean - End-User: 4.40 (Strongly Agree); Expert: 4.65 (Strongly Agree). Analysis: End-users gave the Compatibility statements a mean of 4.40, while IT experts gave a mean of 4.65. For end-users, Pair 2 received the highest mean rating of 4.44, while Pair 5 received the lowest mean of 4.33. For experts, Pair 3 and Pair 5 received the highest mean rating of 4.70, while Pair 1 and Pair 4 received the lowest mean of 4.60. The expert mean was 0.25 points higher than the end-user mean.

Interpretation: Both groups interpreted the Compatibility characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.4 Usability

Table 8. Usability

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	The Klir dashboard or mobile application is easy to understand.	4.35	0.87	Strongly Agree
1	Expert	The user interface follows established usability standards and design principles appropriate to the system's intended users.	4.65	0.67	Strongly Agree
2	End-User	It is easy to find the Klir features I need, such as alerts, reports, monitoring information, or cleaning tasks.	4.38	0.86	Strongly Agree
2	Expert	Navigation elements are logically structured and consistent across the web dashboard and mobile application.	4.65	0.59	Strongly Agree
3	End-User	When an error or warning occurs in Klir, I understand what I need to do next.	4.28	0.95	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
3	Expert	Error messages, warnings, and system feedback provide clear guidance for appropriate corrective actions.	4.70	0.57	Strongly Agree
4	End-User	Klir helps prevent mistakes when completing my tasks.	4.28	0.90	Strongly Agree
4	Expert	The system includes controls that manage user input errors.	4.60	0.68	Strongly Agree
5	End-User	Klir helps me complete my tasks easily and efficiently.	4.41	0.87	Strongly Agree
5	Expert	The interface design promotes efficient completion of monitoring, sanitation, and maintenance tasks.	4.70	0.57	Strongly Agree

Category Mean - End-User: 4.34 (Strongly Agree); Expert: 4.66 (Strongly Agree). Analysis: End-users gave the Usability statements a mean of 4.34, while IT experts gave a mean of 4.66. For end-users, Pair 5 received the highest mean rating of 4.41, while Pair 3 and Pair 4 received the lowest mean of 4.28. For experts, Pair 3 and Pair 5 received the highest mean rating of 4.70, while Pair 4 received the lowest mean of 4.60. The expert mean was 0.32 points higher than the end-user mean.

Interpretation: Both groups interpreted the Usability characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average

rating than the end-user group.

4.4.5 Reliability

Table 9. Reliability

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	Klir works reliably whenever I use it.	4.40	0.79	Strongly Agree
1	Expert	The system operates consistently without unexpected failures during normal operation.	4.60	0.68	Strongly Agree
2	End-User	Klir remains stable during extended periods of use.	4.35	0.83	Strongly Agree
2	Expert	The system maintains its required functionality during prolonged and continuous monitoring operation.	4.60	0.68	Strongly Agree
3	End-User	Klir recovers quickly and continues working when a connection or communication issue occurs.	4.36	0.77	Strongly Agree
3	Expert	Recovery mechanisms, including Wi-Fi and MQTT reconnection logic, successfully restore system operation after connectivity failures.	4.65	0.67	Strongly Agree
4	End-User	My Klir data and records are preserved even when a problem or temporary outage occurs.	4.41	0.79	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
4	Expert	Data persistence and event-queuing mechanisms preserve system records and prevent data loss during temporary outages or failures.	4.65	0.59	Strongly Agree
5	End-User	Klir is accessible and working whenever I need it.	4.36	0.83	Strongly Agree
5	Expert	The system achieves acceptable availability levels for continuous restroom sanitation and facility monitoring.	4.60	0.68	Strongly Agree

Category Mean - End-User: 4.38 (Strongly Agree); Expert: 4.62 (Strongly Agree). Analysis: End-users gave the Reliability statements a mean of 4.38, while IT experts gave a mean of 4.62. For end-users, Pair 4 received the highest mean rating of 4.41, while Pair 2 received the lowest mean of 4.35. For experts, Pair 3 and Pair 4 received the highest mean rating of 4.65, while Pair 1 and Pair 2 and Pair 5 received the lowest mean of 4.60. The expert mean was 0.24 points higher than the end-user mean.

Interpretation: Both groups interpreted the Reliability characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.6 Security

Table 10. Security

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	I feel that my Klir account is secure when I use the dashboard or mobile application.	4.35	0.80	Strongly Agree
1	Expert	User authentication mechanisms effectively prevent unauthorized access to the system.	4.60	0.68	Strongly Agree
2	End-User	My personal information and Klir activity data are protected by the system.	4.38	0.80	Strongly Agree
2	Expert	The system protects sensitive user, sensor, task, and operational data through appropriate security controls.	4.70	0.57	Strongly Agree
3	End-User	I can access only the Klir features and information appropriate to my assigned role.	4.46	0.76	Strongly Agree
3	Expert	Access permissions are correctly enforced according to the assigned user role: Administrator, Supervisor, or Maintenance Personnel.	4.70	0.57	Strongly Agree
4	End-User	I am confident that important actions performed in Klir are properly recorded and tracked.	4.41	0.85	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
4	Expert	The system records and monitors security-relevant activities, including logins, role-based actions, and manual overrides.	4.60	0.68	Strongly Agree
5	End-User	I trust Klir to keep my data safe while it is transmitted and stored.	4.46	0.76	Strongly Agree
5	Expert	Data transmission and storage are adequately protected from unauthorized access through appropriate security measures.	4.65	0.59	Strongly Agree

Category Mean - End-User: 4.41 (Strongly Agree); Expert: 4.65 (Strongly Agree). Analysis: End-users gave the Security statements a mean of 4.41, while IT experts gave a mean of 4.65. For end-users, Pair 3 and Pair 5 received the highest mean rating of 4.46, while Pair 1 received the lowest mean of 4.35. For experts, Pair 2 and Pair 3 received the highest mean rating of 4.70, while Pair 1 and Pair 4 received the lowest mean of 4.60. The expert mean was 0.24 points higher than the end-user mean.

Interpretation: Both groups interpreted the Security characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.7 Maintainability

Table 11. Maintainability

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	Improvements and updates to Klir are implemented effectively.	4.39	0.80	Strongly Agree
1	Expert	The system's modular architecture supports efficient modification, correction, and enhancement of firmware, backend, web, and mobile components.	4.65	0.59	Strongly Agree
2	End-User	Klir updates rarely disrupt my work or normal use of the system.	4.35	0.81	Strongly Agree
2	Expert	The system can be modified efficiently without causing excessive impact on existing functionality.	4.60	0.68	Strongly Agree
3	End-User	Problems I encounter or report in Klir are resolved promptly.	4.36	0.77	Strongly Agree
3	Expert	System faults or defects can be identified, diagnosed, and corrected with reasonable effort.	4.65	0.59	Strongly Agree
4	End-User	New Klir features can be added without affecting the existing features I already use.	4.35	0.73	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
4	Expert	System components are sufficiently modular and reusable to support the addition or replacement of features without adversely affecting existing functions.	4.65	0.67	Strongly Agree
5	End-User	Klir remains dependable during updates and maintenance activities.	4.41	0.77	Strongly Agree
5	Expert	Maintenance activities can be performed with minimal impact on normal system operations.	4.70	0.57	Strongly Agree

Category Mean - End-User: 4.37 (Strongly Agree); Expert: 4.65 (Strongly Agree). Analysis: End-users gave the Maintainability statements a mean of 4.37, while IT experts gave a mean of 4.65. For end-users, Pair 5 received the highest mean rating of 4.41, while Pair 2 and Pair 4 received the lowest mean of 4.35. For experts, Pair 5 received the highest mean rating of 4.70, while Pair 2 received the lowest mean of 4.60. The expert mean was 0.28 points higher than the end-user mean.

Interpretation: Both groups interpreted the Maintainability characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.4.8 Portability

Table 12. Portability

Pair	Respondent	Statement	Mean	SD	VI
1	End-User	I can access Klir from the different supported devices I normally use.	4.34	0.78	Strongly Agree
1	Expert	The system can be deployed and accessed across its supported platforms, including web browsers and Android devices, with minimal modification.	4.65	0.59	Strongly Agree
2	End-User	Klir is easy to access and set up on a supported device.	4.46	0.76	Strongly Agree
2	Expert	Installation, account setup, and access procedures are straightforward and efficient.	4.70	0.57	Strongly Agree
3	End-User	Klir behaves consistently regardless of the supported device I use.	4.39	0.79	Strongly Agree
3	Expert	The system operates consistently across supported browsers, Android devices, and network environments.	4.70	0.57	Strongly Agree
4	End-User	I can use Klir without needing unnecessary special software or hardware beyond the supported requirements.	4.45	0.74	Strongly Agree

Pair	Respondent	Statement	Mean	SD	VI
4	Expert	Platform-specific dependencies are minimized so that the system does not require unnecessary special software or hardware for supported use.	4.60	0.60	Strongly Agree
5	End-User	I can continue using Klir smoothly when I change to another supported device.	4.36	0.77	Strongly Agree
5	Expert	System migration or access from a new supported device can be performed efficiently without loss of user or operational data.	4.70	0.57	Strongly Agree

Category Mean - End-User: 4.40 (Strongly Agree); Expert: 4.67 (Strongly Agree). Analysis: End-users gave the Portability statements a mean of 4.40, while IT experts gave a mean of 4.67. For end-users, Pair 2 received the highest mean rating of 4.46, while Pair 1 received the lowest mean of 4.34. For experts, Pair 2 and Pair 3 and Pair 5 received the highest mean rating of 4.70, while Pair 4 received the lowest mean of 4.60. The expert mean was 0.27 points higher than the end-user mean.

Interpretation: Both groups interpreted the Portability characteristic as Strongly Agree. The results indicate that the statements under this characteristic were rated positively by both respondent groups, with the expert group reporting a higher average rating than the end-user group.

4.5 Summary of Results per Quality Characteristic

Table 13. Summary of Mean Ratings per ISO/IEC 25010 Characteristic

ISO/IEC 25010 Characteristic	End-User Mean	VI	Expert Mean	VI	Overall Mean	Rank
1. Functional Suitability	4.36	Strongly Agree	4.62	Strongly Agree	4.41	7
2. Performance Efficiency	4.37	Strongly Agree	4.65	Strongly Agree	4.42	6
3. Compatibility	4.40	Strongly Agree	4.65	Strongly Agree	4.45	3
4. Usability	4.34	Strongly Agree	4.66	Strongly Agree	4.40	8
5. Reliability	4.38	Strongly Agree	4.62	Strongly Agree	4.43	5

ISO/IEC 25010 Characteristic	End-User Mean	VI	Expert Mean	VI	Overall Mean	Rank
6. Security	4.41	Strongly Agree	4.65	Strongly Agree	4.46	1
7. Maintainability	4.37	Strongly Agree	4.65	Strongly Agree	4.43	4
8. Portability	4.40	Strongly Agree	4.67	Strongly Agree	4.45	2
Grand Mean	4.38	Strongly Agree	4.65	Strongly Agree	4.43	

Analysis: The End-User group recorded its highest characteristic mean for Security at 4.41 and its lowest for Usability at 4.34. The IT Expert group recorded its highest characteristic mean for Portability at 4.67 and its lowest for Functional Suitability and Reliability, both at 4.62 and 4.62, respectively. Across all 100 respondents, Security had the highest overall characteristic mean at 4.46, while Usability had the lowest at 4.40. The grand means were 4.38 for End-Users, 4.65 for IT Experts, and 4.43 when all questionnaire ratings were combined.

Interpretation: All eight characteristics fall within the Strongly Agree range for both groups

and for the combined data. The close spacing among the overall characteristic means indicates that the evaluations were consistently high across the ISO/IEC 25010 dimensions, while the ranking shows the relative differences observed in the combined responses.

4.6 Correlation Between End-User and Expert Evaluations

Table 14. Correlation of End-User and Expert Mean Ratings

Test	Pearson r	Strength	Interpretation
End-User vs. Expert (8 characteristic means)	0.22	Weak	Weak positive relationship

Analysis and Interpretation: The Pearson correlation coefficient between the eight End-User and Expert characteristic means was r = 0.22, which falls within the 0.20 - 0.39 range classified as Weak in Table 4. The result indicates a weak positive relationship between the two groups' characteristic-level ratings. Although both groups gave high ratings overall, their characteristic means did not move together strongly across the eight dimensions.

4.7 Reliability of the Questionnaires (Cronbach's Alpha)

Table 15. Cronbach's Alpha per Characteristic

ISO/IEC 25010 Characteristic	Items	End-User α	Interpretation	Expert α	Interpretation
1. Functional Suitability	5	.950	Excellent	.965	Excellent

ISO/IEC 25010 Characteristic	Items	End-User α	Interpretation	Expert α	Interpretation
2. Performance Efficiency	5	.921	Excellent	.985	Excellent
3. Compatibility	5	.942	Excellent	.985	Excellent
4. Usability	5	.948	Excellent	.983	Excellent
5. Reliability	5	.950	Excellent	.990	Excellent
6. Security	5	.948	Excellent	.984	Excellent
7. Maintainability	5	.933	Excellent	.977	Excellent
8. Portability	5	.940	Excellent	.985	Excellent
Overall Questionnaire	40	.991	Excellent	.998	Excellent

Analysis and Interpretation: All eight characteristics produced Cronbach's alpha values above 0.90 for both respondent groups, which Table 3 classifies as Excellent internal consistency. The overall End-User questionnaire alpha was .991, and the overall Expert questionnaire alpha was .998. These values indicate high internal consistency; however, the CSV also shows predominantly high ratings and limited score variation, especially among the expert respondents.

4.8 Overall Discussion

The evaluation of Klir under the ISO/IEC 25010 software quality model produced a grand mean of 4.43, interpreted as Strongly Agree. End-users recorded an overall mean of 4.38, while IT experts recorded 4.65. All eight quality characteristics were also rated within the Strongly Agree range by both groups. Security received the highest combined characteristic mean at 4.46, while

Usability received the lowest at 4.40. These differences are relative because all eight characteristics received high ratings.

The Pearson correlation between the eight End-User and Expert characteristic means was r = 0.22, classified as a Weak positive relationship using the scale in Table 4. Although both groups gave high overall ratings, their characteristic-level alignment was weak. Overall Cronbach's alpha values were .991 for End-Users and .998 for Experts, both classified as Excellent; the high ratings and limited variation should still be considered when interpreting the reliability results.

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the summary of findings, the conclusions drawn from them, and the recommendations for future researchers and developers of Klir.

5.1 Summary of Findings

The study evaluated Klir using the ISO/IEC 25010 software quality model. Data were gathered from 100 respondents, composed of 20 IT experts and 80 end-users. The questionnaire recorded the end-user group as Student/Maintenance Team and the expert group as Faculty/IT Expert. The findings are as follows:

1. Respondent profile. The dataset contains 80 end-user responses and 20 IT expert responses, giving a total of 100 complete responses with consent. Separate student and janitor counts and expert specialization counts were not collected in the administered questionnaire.
2. Functional Suitability. End-users rated the characteristic Strongly Agree ($\bar{x} = 4.36$) and experts rated it Strongly Agree ($\bar{x} = 4.62$). End-user highest rating: Pair(s) 4 at 4.42; end-user lowest: Pair(s) 1 at 4.31. Expert highest rating: Pair(s) 5 at 4.70; expert lowest: Pair(s) 3 at 4.55.
3. Performance Efficiency. End-users rated the characteristic Strongly Agree ($\bar{x} = 4.37$) and experts rated it Strongly Agree ($\bar{x} = 4.65$). End-user highest rating: Pair(s) 1 at 4.41; end-user lowest: Pair(s) 2, 4 at 4.34. Expert highest rating: Pair(s) 3, 4 at 4.70; expert lowest: Pair(s) 1, 2 at 4.60.
4. Compatibility. End-users rated the characteristic Strongly Agree ($\bar{x} = 4.40$) and experts rated it Strongly Agree ($\bar{x} = 4.65$). End-user highest rating: Pair(s) 2 at 4.44; end-user lowest: Pair(s) 5 at 4.33. Expert highest rating: Pair(s) 3, 5 at 4.70; expert lowest: Pair(s) 1, 4 at 4.60.
5. Usability. End-users rated the characteristic Strongly Agree ($\bar{x} = 4.34$) and experts rated it Strongly Agree ($\bar{x} = 4.66$). End-user highest rating: Pair(s) 5 at 4.41; end-user lowest: Pair(s)

3, 4 at 4.28. Expert highest rating: Pair(s) 3, 5 at 4.70; expert lowest: Pair(s) 4 at 4.60.

6. Reliability. End-users rated the characteristic Strongly Agree ($\bar{x}$ = 4.38) and experts rated it Strongly Agree ($\bar{x}$ = 4.62). End-user highest rating: Pair(s) 4 at 4.41; end-user lowest: Pair(s) 2 at 4.35. Expert highest rating: Pair(s) 3, 4 at 4.65; expert lowest: Pair(s) 1, 2, 5 at 4.60.
7. Security. End-users rated the characteristic Strongly Agree ($\bar{x}$ = 4.41) and experts rated it Strongly Agree ($\bar{x}$ = 4.65). End-user highest rating: Pair(s) 3, 5 at 4.46; end-user lowest: Pair(s) 1 at 4.35. Expert highest rating: Pair(s) 2, 3 at 4.70; expert lowest: Pair(s) 1, 4 at 4.60.
8. Maintainability. End-users rated the characteristic Strongly Agree ($\bar{x}$ = 4.37) and experts rated it Strongly Agree ($\bar{x}$ = 4.65). End-user highest rating: Pair(s) 5 at 4.41; end-user lowest: Pair(s) 2, 4 at 4.35. Expert highest rating: Pair(s) 5 at 4.70; expert lowest: Pair(s) 2 at 4.60.
9. Portability. End-users rated the characteristic Strongly Agree ($\bar{x}$ = 4.40) and experts rated it Strongly Agree ($\bar{x}$ = 4.67). End-user highest rating: Pair(s) 2 at 4.46; end-user lowest: Pair(s) 1 at 4.34. Expert highest rating: Pair(s) 2, 3, 5 at 4.70; expert lowest: Pair(s) 4 at 4.60.
10. Overall rating. The End-User grand mean was 4.38 (Strongly Agree), while the Expert grand mean was 4.65 (Strongly Agree). The combined grand mean across all 100 respondents was 4.43 (Strongly Agree). The highest combined characteristic was Security at 4.46, and the lowest was Usability at 4.40.
11. Correlation. The computed Pearson correlation between the eight End-User and Expert characteristic means was r = 0.22, indicating a Weak positive relationship.
12. Instrument reliability. Cronbach's alpha was α = .991 for the End-User questionnaire and α = .998 for the Expert questionnaire. Both overall values are classified as Excellent internal consistency using the interpretation in Table 3. All eight characteristic-level alpha values for both groups were also above 0.90.

5.2 Conclusion

The conclusions were drawn using the following basis. Each questionnaire statement was rated on a five-point scale and interpreted using Table 2. A statement or characteristic was considered achieved when its mean reached 3.41 (Agree) or higher, and fully achieved when it reached 4.21 (Strongly Agree). Each objective stated in Chapter 1 was matched to the ISO/IEC 25010 statements that evaluate it, and the End-User and Expert means for those statements were used as evidence. The combined mean of 4.43 is the mean of all 100 respondents' ratings (80 End-Users and 20 IT Experts), which equals $0.80 \times 4.38 + 0.20 \times 4.65$.

General objective. The general objective was to design, develop, and validate an IoT-based toilet sanitation and maintenance management system that integrates automated restroom hardware, real-time monitoring, cloud-based task management, and mobile custodial workflows. All eight ISO/IEC 25010 characteristics were rated Strongly Agree by both groups (End-User $\bar{x}$ = 4.38, Expert $\bar{x}$ = 4.65, combined $\bar{x}$ = 4.43). Based on this rule, the general objective was achieved as perceived by the evaluators

Objectives 1, 2, and 4 (occupancy detection, automated lid and flushing sequence, and UV-C disinfection cycle). Functional Suitability Statement 1 evaluates these features directly. It was rated 4.31 by End-Users and 4.60 by Experts, both Strongly Agree. These objectives were achieved.

Objective 3 (water flow and usage telemetry sent to the cloud). Functional Suitability Statement 2 was rated 4.34 by End-Users and 4.65 by Experts, and Compatibility Statement 1 was rated 4.42 and 4.60. All are Strongly Agree. This objective was achieved.

Objective 5 (web-based dashboard). Functional Suitability Statement 4 was rated 4.42 by End-Users and 4.60 by Experts, and Performance Efficiency Statement 4 was rated 4.34 and

4.70. All are Strongly Agree. This objective was achieved

Objective 6 (automated rules and role-based maintenance workflow). Functional Suitability Statement 3 was rated 4.36 by End-Users and 4.55 by Experts, and Security Statement 3 was rated 4.46 and 4.70. All are Strongly Agree. This objective was achieved. The Expert rating of 4.55 is the lowest among all Expert statements, although it remains in the Strongly Agree range.

Objective 7 (React Native mobile application). Usability (End-User $\bar{x}$ = 4.34, Expert $\bar{x}$ = 4.66), Portability (4.40 and 4.67), and Security Statement 1 (4.35 and 4.60) were all rated Strongly Agree. This objective was achieved.

Limits of the evidence. The questionnaires measure perceived quality, not measured hardware performance, which is supported by the test results reported in Chapter 3. The Pearson correlation of r = 0.22 is weak because both groups rated all characteristics almost equally high (End-User means from 4.34 to 4.41 and Expert means from 4.62 to 4.67), which leaves little variation for the correlation to detect. Nineteen of the 80 End-Users and 10 of the 20 Experts rated every item 5, which partly explains the very high Cronbach's alpha values (.991 and .998). The respondents were selected through purposive sampling, and the hardware was validated on a single test bench unit. The conclusions are therefore limited to the information collected by the questionnaires administered.

5.3 Recommendations

Based on the findings and the documented system limitations, the following are recommended to future researchers and developers. Each recommendation states the finding or limitation it is based on.

1. **Deploy more physical units**. The hardware was validated on a single test bench unit. Future

work may install units across more restrooms and floors of the SDCA Annex to test the system at building scale.

2. **Conduct laboratory validation and improve UV-C safety and effectiveness** Since the system now incorporates a true UV-C source, future researchers are recommended to conduct laboratory validation of its sanitation effectiveness through microbial testing, such as colony-forming unit (CFU) measurements, before and after UV-C exposure. The system should also incorporate safety interlocks, occupancy detection, automatic shutoff, and appropriate shielding to prevent accidental human exposure. Further testing should evaluate UV-C exposure duration, coverage, distance, and performance under actual restroom conditions to establish reliable operating parameters. **Basis:** Functional Suitability Statement 1, which covers UV-C disinfection, received the lowest End-User mean in that characteristic (4.31), and the questionnaire did not measure sanitation effectiveness.
3. **Add predictive maintenance with machine learning.** The current design uses threshold and cycle-count rules, Future researchers may train models on the accumulated sensor and task data to forecast component failures, leaks, and cleaning demand. **Basis:** Objective 6 was implemented with rule-based automation, which reacts to events but does not forecast them.
4. **Extend platform support**. The mobile application currently targets Android. An iOS version or a progressive web app would reach more users and improve portability. **Basis:** The mobile application is Android-only, and Experts gave their lowest Portability rating to Statement 4 on platform-specific dependencies (4.60).
5. **Add air quality and consumables sensing.** Odor or air-quality sensors, soap and tissue level sensors, and occupancy counters would let the system trigger cleaning and restocking based on actual conditions. **Basis:** This is based on a limitation of the current design, which triggers

cleaning from flush counts and hardware faults only. It was not measured in the survey.

6. **Add power resilience.** A battery backup or solar-assisted power supply would keep the unit running during outages. **Basis:** This is based on a limitation of the current design. It was not measured in the survey.
7. **Improve the experience for non-technical users.** Consider a Filipino-language interface option, larger touch targets, and short in-app tutorials for custodial staff. **Basis:** End-Users gave their lowest Usability ratings to Statement 3 on understanding what to do after an error or warning and Statement 4 on preventing mistakes (both 4.28).
8. **Strengthen security and audit features.** Multi-factor authentication for administrators and an exportable security audit trail can be added. **Basis:** End-Users gave their lowest Security rating to Statement 1 on account security (4.35), and Experts gave their lowest ratings to Statement 1 on authentication and Statement 4 on activity logging (both 4.60).
9. **Evaluate over a longer period.** A longer field trial with more respondents and a wider mix of campus stakeholders would provide additional evidence of long-term reliability and user adoption. **Basis:** The study used purposive sampling and a short evaluation, and the ratings showed very little variation, so the results cannot be generalized beyond these respondents.

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